

Project Summary

The overall goal of this project is to develop new statistical methods that address important problems in genomics, to implement them in user-friendly open source software, and to make them available to scientists to facilitate new biological discoveries. To this end, the proposed work tackles three important problems arising in genomics where existing statistical methods are lacking, and where new, improved methods could accelerate the pace of scientific discovery: *fine-mapping of functional traits*; *gene set enrichment analysis (GSEA)*; and *discovering overlapping cluster structure in genomic data*.

The work on fine-mapping will enable the identification of genetic variants influencing common sequencing assays such as ATAC-seq and ChIP-seq, without pre-specifying the locations of potential effects. This unbiased approach will help identify regulatory genetic variants and interacting parts of the regulatory genome.

The work on GSEA will provide a new more effective set of tools for researchers who use GSEA to set new findings in the context of known biology. The proposed work uses recently-developed statistical techniques to substantially reduce the redundancy of enriched gene sets, and will provide researchers with succinct and precise results that better highlight the full range of known biological factors that are relevant to a new set of findings.

The work on overlapping cluster structure will provide new generally-applicable methods for understanding complex layered and hierarchical relationships that occur commonly in genomics applications (e.g. cell subtypes nested within cell types, layered on top of patient effects).

The tools developed here will help scientists tackle a diverse range of analysis problems that arise in genomics, ultimately helping them better understand the biology of disease, with the eventual goal of improving therapies and treatment strategies.

Project Narrative

Public health relevance: This project will generate and apply statistical tools for analyzing large-scale genomic studies. The tools will help scientists tackle a diverse range of problems, including understanding the effects of genetic variation on fundamental biological processes, identifying collections of related genes that may help inform or explain new experimental results, and discovering relationships and clustering among samples (e.g., cells, genes or people), for example to identify cell or disease sub-types. Solving these problems has the potential to help understand the biology of disease, eventually leading to new therapies or treatment strategies.

Budget Justification

Stephens, Matthew, Ph.D., Principal Investigator, (4 months effort) will be responsible for the overall direction of the project and supervision of supporting research personnel. He will undertake research with regard to all specific aims. Dr Stephens is an international leader in the field of statistical genetics, and has a strong record of developing novel statistical methods for analyzing large-scale genomic datasets, and of distributing user-friendly open source software implementing them. The recent major software contributions of his lab include the *susieR* package for fine-mapping, the *ashr* and *mashr* packages for multiple testing, and *flashr* for sparse matrix factorization. The proposed work builds on these strong foundations to generate new and important methods for several important problems in genomics.

Dr William Denault, Postdoctoral Researcher (12 months effort) will lead research related to Aim 1, and also contribute to Aim 2. Dr Denault is a talented statistical scientist, with an MS in Statistics and PhD in Public Health and Genetic Epidemiology. He has extensive experience developing methods for functional traits, and his PhD thesis was "Wavelet-based methods in genetic association analysis." Dr Denault helped prepare preliminary data for Aim 1, and he is ideally qualified to lead this work.

Karl Tayeb, Graduate Student Researcher (12 months effort) will undertake research related to Aim 2. Mr Tayeb is an outstanding graduate student with exceptional computational and software skills, and strong undergraduate research experience. He helped to prepare preliminary data for Aim 2, and is ideally placed to continue this work.

Postdoctoral Researcher, to be appointed (6 months effort) We will appoint a postdoctoral scholar with experience in computational and statistical genetics to help lead the proposed research in Aim 3.

Peter Carbonetto, Ph.D., Computational Scientist (6 months effort) will help lead work on Aim 3, and also contribute to all Aims. Dr Carbonetto is an experienced computational researcher, with a PhD in computer science, and postdoctoral training in computational and statistical genetics. He has extensive experience in software development Best Practices, as well as being an expert in computational methods, particularly variational Bayes and Markov chain Monte Carlo. Dr Carbonetto is the author of the *fastTopics* software package for topic modelling of single-cell data, so he is well placed to contribute to Aim 3. In addition, his role on the project includes assisting the postdoctoral researchers to help ensure that their research is rigorous and reproducible, using software development Best Practices (e.g., version control and unit tests), and helping to document and test software for distribution to the community. PI Stephens will meet with Dr Carbonetto at least biweekly to discuss progress.

Introduction to Revised Application:. We thank the reviewers for their helpful comments on the original proposal, which we have worked to fully address in this revised application. We were pleased that the original proposal was very well received (Impact score 31; percentile 14), but disappointed that this fell below the pay line. We hope reviewers will agree that our revision addresses the weaknesses noted and that the revised proposal is still more highly significant, innovative, and impactful.

Response to Summary Statement:. *A larger part of the reviewers saw the project as highly significant and innovative... the project tackles three key methodological challenges A strong plan for dissemination via software...Rigor of the prior research evaluation is strong...[it will] yield statistical methods and software broadly applicable in population genetics and single-cell genomics research. The other reviewer group saw the project as moderately significant and innovative...proposed methods build heavily on existing ones...the data to be analyzed are not detailed... the investigator team does not include disease-specific expertise for the data-analysis result interpretation.* Naturally we disagree with the latter reviewer group, and were pleased that the larger part of the reviewers appreciated the high significance, innovation and impact of our proposed work. The reason for this high impact is that we will *develop and disseminate software implementing widely applicable new statistical methods for important problems.* While we also aim to conduct interesting and appropriate data analyses, ultimately its high impact comes from the *multiplicative factor* that comes from repeated use of our methods by scientists around the world.

Modifications and responses to specific comments:.

A major weakness ...is that applicability and interpretation of the Aim 3 methods to single-cell genomics analysis is undefined, and lacks preliminary data. We have added extensive new preliminary data to demonstrate the high potential for our methods to provide new insights in single-cell data. We emphasize that our methods are very general, and could apply to any applications where clustering and NMF are currently applied.

Another weakness...is not accounting for biological variables in the data-analysis plans... Sample sizes of the available studies and the kinds of features available from various studies are not explained. The gender distribution of the available studies is unclear. Details are necessarily brief due to space limitations, but we have improved our methods development descriptions and data analysis plans to clarify that they will account for biological variables and other covariates. We have added sample size and gender distribution.

The study lacks co-investigators or collaborators having disease-specific expertise such as expertise in pancreatic ductal adenocarcinoma (for Aim 3) We have engaged with the Pancreas Working Group at the University of Chicago, including PDAC experts Professors K. Macleod and S. Oakes. We have presented and discussed our preliminary results with them, and they have helped us to link them with known PDAC biology. They will continue to help us interpret future results (see letter of support). If results are sufficiently promising this could lead to follow-up experiments, but this would require a separate funding application.

The study also does not appear to have access to novel data sets as Aims 1 and 2 rely entirely on data published in other studies. We do not believe that analysis of unpublished data is necessary to accomplish the primary goals of this proposal. Papers describing high-impact methods (e.g. ADMIXTURE, EIGENSTRAT, PHASE, BEAGLE, IMPUTE, Mach, GEMMA, boltLMM, SuSiE, CAVIAR, FINEMAP, *t*-SNE, UMAP, qvalue, ...) typically do not analyze unpublished data. We hope that reviewers will, on reflection, agree that using published data to aid methods development is appropriate and does not reduce the impact or significance of our proposal.

The study lacks methodological innovation...methods are straightforward expansions of known methods – for example, Aim 1 appears to be a direct extension of SuSiE, Aim 2 appears to be a straightforward application of logistic regression or standard mixture modeling of effects with latent variables, and Aim 3 appears to be standard matrix factorization. We fundamentally disagree. Our proposed methods build on existing work, but every aim involves a *novel combination of ideas.* Putting existing ideas together in new ways to achieve new and important things is innovation. Aim 1 will provide the *first available methods* for fine-mapping of functional traits. Aim 2 aims to *change the way the field thinks about GSEA* (marginal tests followed by post-hoc clustering); and the mixture model is not “standard” because it uses a recently-invented variable selection prior to link the mixture proportions to many correlated annotations, something no existing method can do. Finally, Aim 3 represents an *entirely new approach*, which although based on matrix factorization, cannot reasonably be described as “standard”. We believe this proposal continues our strong tradition of developing innovative and impactful methods for genomics applications.

Specific Aims

The overall goal of this project is to develop new statistical methods that address important problems in genomics, to implement them in user-friendly open source software, and to make them available to scientists to facilitate biological discoveries. To this end, we have identified three important problems arising in genomics where existing statistical methods are lacking, and where new, improved methods could accelerate the pace of scientific discovery: *fine-mapping of functional traits*; *gene set enrichment analysis* (GSEA); and *discovering overlapping cluster structure in genomic data*. Upon completion, the methods we develop here have the potential to **change the way that several fundamental analysis challenges are tackled in genomics**.

Our proposed work exploits recent advances we have made in two key areas of statistical methodology: a new approach to variable selection in regression called *SuSiE* [1, 2]; and a new approach to matrix factorization called empirical Bayes matrix factorization (EBMF) [3]. Although published only in 2020, *SuSiE* has already established itself as a state-of-the-art approach to fine-mapping, and our most used software package: it has 115 stars on Github, more than any of our other packages, and >12,500 downloads in 1 year from CRAN. We see two key reasons for this popularity: (a) it is fast, and (b) it provides a simple summary of uncertainty in variables selected using “credible sets” (CSs). Here we will exploit these unique features to produce new fine-mapping methods for functional traits (e.g., as measured by high-throughput sequencing assays), and a whole new approach to GSEA that we believe will change the way GSEA is done in genomics. Our work on EBMF has also found widespread use (software *flashr* has 46 stars on Github) and our proposed work here will extend its capabilities for identifying *overlapping cluster-like structures* that occur in many genomic applications, ranging from human population genetics to single-cell transcriptomics to mutation signature analysis.

Aim 1: Develop and apply new statistical methods for fine-mapping of functional traits. Building on *SuSiE*, these new methods will be the first to allow joint fine-mapping of molecular traits measured by sequencing assays such as ATAC-seq and ChIP-seq, without pre-specifying the locations of potential effects. This unbiased approach will help identify regulatory genetic variants and potentially interacting parts of the regulatory genome.

Aim 2: Develop new statistical methods for joint Gene Set Enrichment Analysis (GSEA). GSEA is one of the frequently-performed analyses in genomics. GSEA helps set new findings in context, by identifying gene sets (pathways, gene ontologies, *etc*) that are overrepresented in a list of genes identified as “interesting” in a new analysis. However, the way GSEA is currently performed, one gene set at a time, tends to produce a plethora of redundant results due to strong overlap among gene sets. Our work here will address this redundancy by using *SuSiE* to *jointly* analyze all gene sets, and to *distinguish redundant enrichments from complimentary enrichments*.

Aim 3: Develop and apply new methods for discovering overlapping cluster structure in genomic data. Overlapping cluster structure is ubiquitous in genomic data, but existing methods for discovering such structure are surprisingly limited. By building on our EBMF framework [3], we propose a new method, empirical Bayes covariance decomposition (GBCD), to better identify such structure. We will apply our methods to a range of applications, including population genetics, single-cell genomics and mutational signature analysis, and compare our methods with existing state-of-the-art methods such as admixture models, topic models, and non-negative matrix factorization (NMF) methods.

This project will produce freely available open source software implementing these methods that will be invaluable to the broad community of genome scientists who use methods for fine-mapping, GSEA, and identification of cluster structure.

The investigator. PI Matthew Stephens is an international leader in the development of statistical and computational methods for genetic data analysis, and a Highly Cited Scientist with over 115,000 total citations, and an *h*-index of 83 (Google Scholar, 6/2022). Stephens has made key contributions to many areas important in this proposal, including methods for: statistical fine-mapping [1, 4, 5]; univariate and multivariate multiple testing [6, 7]; and dimensionality reduction, clustering and principal components analysis [3, 8–12]. He is ideally positioned to lead the proposed work.

Research Strategy

Progress Report. This is a proposed renewal of PI Stephens's primary R01, which supports statistical methods development work in the Stephens lab. This progress report covers period 7/1/2018–6/30/2022 (currently in NCE). The overarching aim of this grant is to develop, apply, and distribute statistical analysis tools for genomic studies. During the current period of support we developed new methods and software for: (i) fine-mapping of single traits [1, 2, software *susieR*] (ii) multivariate multiple testing [7, software *mashr*]; and (iii) matrix factorization and dimensionality reduction methods based on Empirical Bayes matrix factorization [3, 13, software *ebnm*, *flashr* and *flashier*] and topic modeling [12, software *fastTopics*]. This period produced 27 publications (21 journal papers and 6 preprints), including 4 in *Nature Genetics*, 3 in *Nature Communications*, 3 in *PLoS Genetics*, 1 in *Biostatistics*, 2 in *Journal of Machine Learning Research*, and 1 in *Journal of the Royal Statistical Society*. Of 27 publications, 19 involve primarily members of the Stephens lab supported by this R01, while 8 are collaborations with other labs. Details on the most relevant work are given as needed in Approach.

A. Significance.

Summary. This project will develop and apply novel statistical methods for three important problems arising in genomics applications: *fine-mapping of functional traits*; *gene set enrichment analysis* (GSEA); and *discovering overlapping cluster structure in genomic data*. The methods we develop here **have the potential to change the way these fundamental analysis challenges are tackled in genomics**.

Fine-mapping and colocalization of molecular functional traits. Fine-mapping is the process of narrowing down a genetic association signal to a set of potential causal variants [14–16]. Fine-mapping is thus key to understanding the relationship between genotype and human disease. Closely related to fine-mapping, *colocalization analysis* aims to determine whether genetic association signals in the same genomic region for different traits are explained by the same causal variant(s) [17–21]. By linking a disease trait with upstream molecular traits, colocalization can help elucidate the *molecular mechanism by which genotype impacts disease risk*. More generally, fine-mapping of molecular traits like gene expression [22], chromatin accessibility [23], methylation [24, 25], transcription-factor binding [26, 27] and histone modifications [28, 29] can help identify regulatory genetic variants that are candidates for impacting organism-level traits relevant to human health.

Underlining its importance, many statistical methods have been developed for fine-mapping [1, 2, 4, 15, 30–39]. However, existing methods analyze only one univariate trait, or perhaps a few quantitative traits [40–42]. Many traits, including molecular measurements from **-seq* assays such as ChIP-seq [43] and ATAC-seq [44], as well as CpG methylation levels measured by whole-genome bisulfite conversion (WGBC) [45], do not naturally fall into this framework. Instead, they are more appropriately considered as *functional traits* that vary along the genome. While methods exist for *association analysis* of functional traits [46–50], these methods test for association with each SNP, and do not distinguish between the causal SNP and those in linkage disequilibrium (LD) with the causal SNP. *Currently, no tool exists for statistical fine-mapping of functional traits*.

Our proposed work in Aim 1 will fill this gap, providing methods and software that can **fine-map molecular traits at higher resolution and with greater power than is currently possible**. By developing methods to fine-map multiple functional traits simultaneously, our work will help identify regions of the genome that are co-regulated and enable new insights into the regulatory landscape of the human genome.

Joint gene set enrichment analyses. GSEA is a widely used tool to distill the results of large-scale genomic analyses into more immediate biological insights. In brief, GSEA assesses whether a predefined group of related genes (“gene set”, GS) is overrepresented, or “enriched”, among a set of “interesting” genes generated from an analysis (“interesting” can be quantified in many ways, e.g., small *p* values or large effect size estimates). GSEA was introduced as a specific method in [51], but many methods now exist [52–74], and we use “GSEA” as a generic term here. GSEA is ubiquitous in genomics studies because it helps researchers interpret results in the context of known biology. The original GSEA paper [51] has been cited >32,000 times, and a commonly used web server (Enrichr [73]) reports processing >48 million queries.

A major limitation of most GSEA analyses is that they are conducted *marginally*: GSs are tested *one at a time* for enrichment. This results in long lists of significant enriched GSs because many GSs are substantially overlapping, or even nested, so enrichment in one GS can drive enrichment in many other similar GSs. To

address this issue, existing GSEA methods offer *post hoc* methods for clustering, summarizing or visualizing redundant enrichment results (e.g., [71–76]). However, this still results in many clusters, each containing many GSs, and *ad hoc* decisions about which GSs to cluster together. Further, GSs in different clusters may still be correlated, so enrichment of one cluster may actually be explained by a GS in another cluster. Other methods for addressing redundancy in GSEA [e.g. 57, 61, 66, 77–79] rely on complex models, or heuristics.

Our proposed work in Aim 2 addresses this significant problem by replacing thousands of *marginal* analyses with a single *joint* analysis of all GSs. We achieve this by framing GSEA in a regression framework [58, 80], and introducing a sparse prior for the enrichment parameters. Our proposal exploits our recently developed *SuSiE* fine-mapping method [1], which is ideally suited to managing highly correlated GSs. Our work will provide a new principled approach to evaluating the relative evidence for enrichment among competing GSs. By managing redundancy, our approach will *identify complimentary enrichments*, ultimately leading to simpler and more precise biological insights from this widely used analysis.

Discovering overlapping cluster structure in genomics data. Cluster-like structures are ubiquitous in genomic data. These discrete structures are often overlapping, and sometimes hierarchical; however, not all the structure is exactly hierarchical or exactly discrete. For example, in **human population genetics**, there is structure at a continental level (e.g., Europe vs. Asia vs. Africa) and also within continents (e.g., between countries). This structure is largely hierarchical, and somewhat discrete. However, the complexity of historic human migrations, intermixing and admixture produces deviations from a straightforward hierarchy. In **single-cell genomics**, analyses often classify cells into distinct “cell types” before attempting to distinguish subtypes. This structure is hierarchical and somewhat discrete. However, many biological processes (e.g., cell cycle, inflammation) are active across cell types, so the data do not conform to a simple hierarchy.

Conventional clustering methods assume that clusters are non-overlapping (every sample belongs to one cluster), or arranged in a simple hierarchy (“hierarchical clustering”). Limitations of conventional clustering methods motivated the development of more flexible models that allow samples to belong to more than one cluster. Prominent examples include admixture models in population genetics [8, 81, 82]; topic models [83], which have recently become widely used in single-cell sequencing [11, 12, 84–90]; and Non-negative matrix factorization (NMF) [91] in mutational signature analysis [92–94]. Admixture models, topic models and NMF are all very widely used (the original papers [8, 83, 91] have >85,000 citations in total). The Stephens lab has contributed extensively to development and application of these methods in the past 20 years [8, 10–12, 95–99]. However, we have recently found that *they often miss the overlapping and hierarchical structures that occur in many genomics data sets* (see Preliminary Studies). We therefore see exciting opportunities for new methods that better elucidate overlapping cluster structure, and we propose a new approach, generalized binary covariance decomposition (GBCD) to achieve this aim.

User-friendly open source software. All three aims tackle problems that arise repeatedly in genomics, so the methods we will develop have the potential for widespread application and impact. We will develop and distribute fully-documented user-friendly open source software to maximize the significance of this work.

B. Innovation. The Stephens lab has a strong record of developing innovative statistical methods for important problems. The proposed work in all three aims continues this tradition of innovation:

1. Our work in Aim 1 will provide the first rigorous fine-mapping methods designed for functional traits. This work exploits the innovative new *SuSiE* framework for fine-mapping introduced in the current funding cycle, whose speed and flexibility make these new approaches possible.

2. Aim 2 proposes an innovative approach to dealing with correlated or competing gene sets in GSEA. Rather than pre-cluster or post-cluster gene sets, our proposed methods treat GSEA as a variable selection problem. This innovative approach exploits the ability of *SuSiE* to produce CSs that *distinguish between redundant variables and complementary variables*: redundant variables (those competing to explain the same association signal) are contained in the same CS, while complementary variables (those that explain independent signals) are contained in different CSs. This innovation has already changed the way fine-mapping results are usually reported, and has the potential to do the same for GSEA.

3. Our proposed GBCE method to discover overlapping cluster structure represents an entirely new ap-

proach to factor analysis for high-dimensional data. This work builds on the EBMF framework introduced in the current funding cycle to model overlapping cluster structures using binary prior distributions. Our preliminary results suggest this can provide new insights into the structure of many types of genomic data.

C. Approach.

Aim 1: Fine-mapping and colocalization of functional traits.

Preliminary Studies: methods for fine-mapping. Although published only in 2020 [1], our new method *SuSiE* has quickly established itself as a state-of-the-art approach to fine-mapping. It has already become widely used in practice (e.g., [100–102]), and has been incorporated into the widely used colocalization tool *coloc* [20]. The attractions of *SuSiE* include its speed, and its ability to assess uncertainty in a simple and intuitive way via *credible sets* [1]. In the current funding period, we developed versions of *SuSiE* for fine-mapping of quantitative traits from both full data [1] and summary data [2], and for fine-mapping multiple related traits [42].

SuSiE, like other Bayesian fine-mapping methods [4, 15, 30–39], relates a trait vector $\mathbf{y}$ to a genotype matrix $\mathbf{X}$ using multiple regression, $\mathbf{y} = \mathbf{X}\mathbf{b} + \mathbf{e}$, with a sparse prior on the vector $\mathbf{b}$. Assuming all SNPs are measured and there is no confounding due to population structure, the coefficient b_j estimates the causal effect of SNP j because multiple regression controls for effects of other SNPs in LD with j . A key innovation behind *SuSiE* is a new way to specify the sparse prior on the vector $\mathbf{b}$ by writing it as a sum, $\mathbf{b} = \sum_{l=1}^L \mathbf{b}^{(l)}$, where each vector $\mathbf{b}^{(l)}$ is assumed to have *exactly one non-zero element* (a “single-effect vector”). Thus, $\mathbf{b}$ is a “Sum of Single Effect” vectors, which gives *SuSiE* its name. (Note that L is a user-specified upper bound on the number of effects in the region; [1] shows *SuSiE* is robust to specification of L provided it is sufficiently large.) This new prior specification allows for a fast and convenient fitting procedure—Iterative Bayesian Stepwise Selection (IBSS)—which iteratively estimates each single-effect vector $\mathbf{b}^{(l)}$ in turn. Estimating a single-effect vector is easy and efficient because it has only one non-zero value, for which there are only p configurations, all of which can be quickly evaluated. [1] shows that IBSS can be rigorously justified as optimizing a variational approximation to the posterior distribution on $\mathbf{b}^{(1)}, \dots, \mathbf{b}^{(L)}$, and the experiments in [1] illustrate its competitive performance compared with existing state-of-the-art fine-mapping methods.

Preliminary Studies: QTLs for *-seq assays and limitations of peak-finding.

Molecular traits measured by *-seq assays can be considered as “functions” that vary along the genome. Similarly, the effect of genetic variants on such traits can also be considered as a function. To illustrate, Figure 1 shows *-seq data from the lab of Dr L Barreiro [103] — including RNA-seq [104], ATAC-seq [105], and ChIP-seq [106] on four histone marks [29] — for a study of monocyte-derived macrophages (MDMs) derived from 35 healthy males, 14 of African American ancestry and 21 of European ancestry. The figure shows a QTL, rs1994292, identified in this region based on association with ATAC-seq data in two “peaks” identified by MACS2 [107, 108]. Visually, the SNP also associates with other traits: the non-reference allele is associated with increased expression, increased levels of chromatin accessibility, H3K4me1, H3K27ac and H3K4me3, and reduced levels of the repressive mark H3K27me3, with effect size varying along the region in different ways for different traits. However, for three histone marks MACS2 identified no peaks, so peak-based analysis detects no associations in these traits. Furthermore, to assess whether the same SNP likely underlies the association signal in all six traits requires joint consideration of all SNPs in the region, and patterns of LD, because associations at one SNP can be driven by LD with another causal SNP. That is, a rigorous analysis requires *new tools for fine-mapping of functional traits*. Our proposed work will provide such tools.

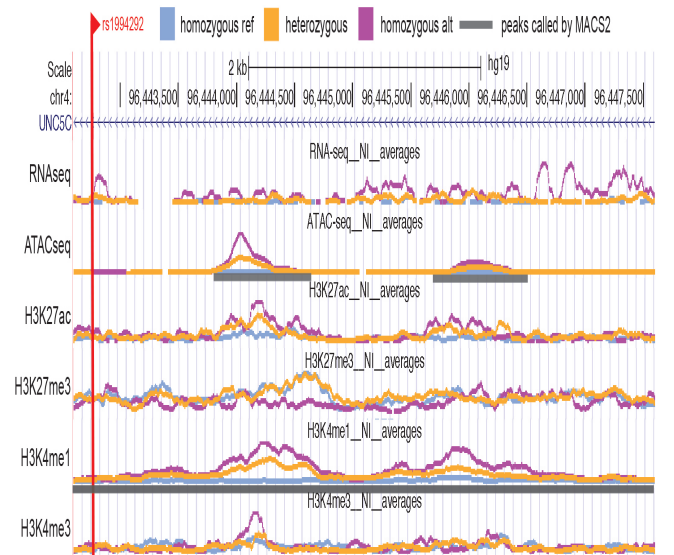


Figure 1: Example QTL for molecular traits measured by *-seq data near gene *UNC5C*. Each colored curve shows average coverage for a different genotype class.

Preliminary studies: association analysis of functional traits. To address limitations of peak-based analyses, several methods have been developed for peak-free association analysis of functional traits. These include methods based on “bump-hunting” [49, 109–112], hidden Markov models [113, 114] and wavelets [47, 48, 50, 115]. These existing methods can assess association between a *single SNP* and a single functional trait, and increase power to detect effects that do not exactly overlap a prespecified peak or window [47].

Proposed work. Here we propose to combine wavelet-based functional association methods with the *SuSiE* prior to produce new powerful methods to *fine-map functional traits*. We will apply these methods to understand connections among genetic variants, histone marks, chromatin accessibility, methylation and expression.

Aim 1a: Functional *SuSiE*, “*fSuSiE*”. In this aim, we will implement a new statistical method to fine-map a single functional trait. Assume we have data on N samples at T locations arranged in a data matrix, $\mathbf{Y} = (y_{it})$. Thus, y_{it} is the observed trait in sample i at location t . For example, if the trait is chromatin openness then Y_{it} could be the number of reads from sample i mapping to location t in a genomic region. To fine-map functional traits we will use multivariate regression, $\mathbf{Y} = \mathbf{XB} + \mathbf{E}$, where $\mathbf{X}$ is an $N \times p$ matrix of SNP genotypes in the region to be fine-mapped; $\mathbf{B}$ is a $p \times T$ matrix whose element b_{jt} is the effect of SNP j on the trait at location t ; and $\mathbf{E}$ is a matrix of independent $N(0, \sigma^2)$ error terms. In fine-mapping the matrix $\mathbf{B}$ is assumed to be sparse—most rows will be zero, corresponding to SNPs that do not affect trait. The goal is to identify the non-zero rows. Furthermore, a key feature of functional traits is that the SNP effects will tend to be “spatially structured”; thus, in any row of $\mathbf{B}$ containing nonzeros, the change $|b_{jt} - b_{j,t+1}|$ will often be small.

Applying the *SuSiE* approach to this setting, we write $\mathbf{B} = \sum_{l=1}^L \mathbf{B}^{(l)}$, where each “single-effect” matrix $\mathbf{B}^{(l)}$ has only one non-zero row, corresponding to a single causal SNP. The matrix $\mathbf{B}$ therefore has at most L non-zero rows. Analogous to univariate *SuSiE* [1], we will fit the model using the IBSS algorithm, which optimizes a variational approximation to the posterior distribution for $\mathbf{B}^{(1)}, \dots, \mathbf{B}^{(L)}$. The IBSS algorithm involves repeatedly fitting *single-SNP models*, which are of the form $\mathbf{Y} = \mathbf{x}_j \mathbf{b}'_j + \mathbf{E}$ where $\mathbf{x}_j$ is a column vector of the SNP genotypes at SNP j , and $\mathbf{b}'_j$ is a row vector of the effects at that single SNP. Specifically, IBSS requires the following computations for each SNP j : (i) the single-SNP Bayes Factor comparing the “alternative” model that $\mathbf{b}_j$ is non-zero against the null model, $\mathbf{b}_j = \mathbf{0}$; (ii) the posterior mean effect size estimate $E(\mathbf{b}_j | \mathbf{Y}, \mathbf{x}_j)$. The wavelet-based methods for functional association analysis in [47] provide exactly these single-SNP computations. Thus, we will implement *fSuSiE* using these methods. These methods use empirical Bayes to automatically adapt to the smoothness (or otherwise) of the underlying functional signals, making them applicable to diverse assays.

Like *SuSiE*, the final output of *fSuSiE* will be (i) a posterior inclusion probability for each SNP, quantifying the probability that it has a non-zero effect; (ii) effect size estimates, and their standard errors, for each SNP at base-pair resolution along the genome, which we can use to identify genomic regions with significant effects; (iii) a list of credible sets (CSs), each designed to contain at least one SNP with non-zero effect (with high probability, e.g., >0.95). The number of CSs reflects the number of independent association signals discovered, and the number of SNPs in each CS reflects the number of SNPs that are plausible candidates for explaining the signal. (Typically, SNPs in the same CS are in very high LD, >0.9 [1].)

Aim 1b: Multivariate *fSuSiE*. Building on Aim 1a, we will extend *fSuSiE* to *jointly* fine-map multiple functional traits. To do this, we will first implement single-SNP association methods for multiple functional traits; once such single-SNP computations are implemented, they can be incorporated into an IBSS algorithm to achieve fine-mapping in the same way as for univariate *SuSiE* and *fSuSiE*. Letting m index the M functional traits, the single-SNP models are of the form $\mathbf{Y}_m = \mathbf{x}_j \mathbf{b}'_{jm} + \mathbf{E}_m$, $m = 1, \dots, M$. The key issue is specification of an appropriate computationally-tractable prior distribution for the effects $\mathbf{b}_{jm}$ across m different traits. We propose to implement two approaches: (i) independent (wavelet-based) priors across the m traits. This has the benefit of simplicity and serves as a starting point for comparing more sophisticated alternatives; (ii) extend the wavelet-based priors from [47] to incorporate correlations across the m traits. Specifically, we will replace the univariate mixture of normal priors for wavelet coefficients used in [47] with the mixtures of multivariate normal priors used in the *mashr* software [7]. This approach will be more computationally intensive, but could improve power to detect overlapping effects where a single SNP affects different traits in the same region.

Extensions. We will extend our methods in Aims 1a and b to adjust for biological variables (e.g. sex) and

covariates (e.g. batch, ancestry), and also to incorporate individual random effects. Because our proposed approaches are built on fitting *single-SNP models* these extensions will simply involve modifying the single-SNP computations to incorporate these features, which is straightforward based on existing single-SNP methods.

Assessments, benchmarks and comparisons. We will conduct extensive assessments and benchmarking of our new methods. First, we will generate realistic “null” data by randomly pairing real **-seq* data with real genotype data. This will allow us to empirically verify false positive rates of all methods. Next, we will modify the null data to “spike-in” QTL signals using binomial thinning [116]. To make the simulations as realistic as possible we will spike in QTL signals estimated from real data. Using these simulations, we will assess power (proportion of causal variants captured by CSs) and coverage (proportion of CSs including a causal variant). We will also assess sensitivity to the choice of L . Since there are no existing methods for fine-mapping functional traits, we will compare peak-based analyses that first identify peaks in the **-seq* data and then fine-map each peak as a univariate trait. Since peak-finding is itself an error-prone process, and some QTL effects may fall outside detected peaks, we anticipate a substantial gain over these approaches.

Aim 1c: Apply methods to fine-mapping of molecular traits. To ensure our methods are tractable and suitable for practical use we will apply them to fine-map QTLs in real data. As a starting point we will use the **-seq* data collected by Dr L Barreiro [103] (samples as described in Preliminary Studies). These data have several features that make them well suited to our purpose: they are very deeply-sequenced (211 billion reads total; on average .5 billion per sample-assay pair), which will help exploit our methods’ potential for high-resolution mapping; they include 6 different assays, which will help fully exploit our multivariate models; they include multiple ancestries, providing an opportunity to assess our methods for controlling for covariates; and they are generated by a lab who are already intimately familiar with the data, have preprocessing, QC and peak-based analysis pipelines in place that we can use and compare with, and who as motivated potential users of our methods will provide feedback on the results and user interface (see letter of support from L Barreiro).

To apply our new *fSuSiE* methods to these data, we will analyze moderate-sized genomic regions (e.g. 500kb) around each gene and use methods developed in Aims 1a,b to identify putative causal SNPs affecting one or more molecular traits. For each causal SNP we will identify the genomic regions affected in each trait. We will compare the results with peak-based analyses from [103]. We will also assess the impact of controlling for population structure and ancestry, as well as introducing individual random effects.

To seek new insights into underlying regulatory mechanisms we will use the *fSuSiE* results to infer: (i) the shared genetic architecture of the molecular traits (which traits tend to share genetic factors, and related questions such as how far apart the regions affected tend to be); (ii) whether genetic variants disrupting transcription factor binding sites have a distinctive signature in their genetic effects; (iii) the extent to which affected regions in different traits are bounded by topologically associating domains (TADs) [117]; (iv) whether genetic fine-mapping can connect enhancer elements with the genes they regulate (e.g., a SNP that affects both expression of a gene and chromatin accessibility in an enhancer suggests a link between enhancer and gene); and (v) whether fine-mapping of regulatory QTLs can contribute to understanding of the gene regulatory mechanisms underlying GWAS hits [118].

Potential pitfalls and alternative approaches.. Our proposed methods represent a novel combination of two existing ideas: *SuSiE* for fine-mapping [1] and wavelet-based association analysis of functional traits [119]. Since both parts have been shown to perform well, we do not anticipate problems with combining them. The second approach in Aim 1b is somewhat ambitious, but the first approach (independent priors) is straightforward, and should outperform the simpler alternative of fine-mapping each trait separately. We will make this comparison.

We chose wavelet-based methods for functional analysis because of their computational efficiency. Nonetheless, we anticipate some computational challenges fine-mapping **-seq* assays in large regions at base-pair resolution. We will explore strategies to reduce computation, including pre-summarizing data in bins of fixed width (e.g., 20bp bins reduces computation by 20-fold), and discarding high-resolution wavelet coefficients. Both approaches would reduce computation at the expense of ability to resolve signals affecting small regions, and we will assess the impact of these approximations.

The proposed methods assume normally-distributed errors, which may seem inappropriate for handling sequencing data. However, [48] compared wavelet-based methods with a multiscale Poisson (msP) approach

tailored to sequence data, and found the wavelet-based methods performed well provided n is not small (e.g., $n > 10$). Since fine-mapping studies involve $n \gg 10$, we propose to start with the simpler wavelet-based methods and assess their adequacy by simulation. If necessary, we will adapt the msP approach to this setting.

The data from [103] have advantages (above) but also limitations, being all male and relatively small sample size. Our methods will be applicable to larger datasets containing both sexes, and when our methods are mature we will assess them to such data. One potential dataset (RNA-seq and ATAC-seq on 400 individuals of both sexes) is currently being collected by the Barreiro lab (letter of support).

Aim 2: New tools for joint GSEA.

Preliminary Studies. Almost all current gene set enrichment analyses (GSEAs) are performed marginally—that is, one gene set at a time. Because gene sets (GSs) are often highly correlated, this process identifies many redundant enriched GSs. To illustrate this problem, we performed GSEA using a list of 1,316 genes found to be differentially expressed in CD19+ B cells vs. other immune cell types (data from [120]). Marginal GSEA (Fisher’s exact test, FDR = 0.05) for 9,896 GSs from the Gene Ontology [121, 122] identified 1,117 enriched GSs. Thus, *more than 10% of all GSs tested were marginally enriched*, and the list of enriched GSs was nearly as long as the original list of genes! This is not a defect specific to Fisher’s exact test, which is widely used for GSEA [69–74]; any marginal method will produce many redundant enrichments because many GSs are indeed *marginally enriched* due to overlap with truly “relevant” GSs. Indeed, as expected, the significant GSs contain many overlapping GSs capturing the same or similar underlying biology. This redundancy obscures the most relevant independent enrichment signals, and encourages the selective reporting of enrichment results.

Existing methods for this problem try to post-process marginal enrichment results to reduce redundancy. To illustrate this, we applied two widely used methods, DAVID [75, 123] and the affinity propagation (AP) method used in WebGestalt [71]. DAVID clustered the 1,117 significant GSs into 538 “annotation clusters,” and WebGestalt returned 235 significant GS clusters containing 1,215 GSs in total. Thus, both methods yielded only a minor reduction in redundancy, and neither provided a succinct summary of the relevant enriched processes.

Proposed Work.

Aim 2a: Joint GSEA via sparse logistic regression. In this aim, we tackle the simplest and most widely used version of GSEA, which analyzes a list of “interesting” genes identified from some known “universe” of n genes (which serves as the “background” against which enrichment is assessed). Given a list of p GSs for analysis, we will create an $n \times p$ binary matrix $\mathbf{X}$, where x_{ij} is the indicator, $x_{ij} = 1$ if gene i is in gene set j , otherwise $x_{ij} = 0$. Let $\mathbf{y}$ denote a binary vector with $y_i = 1$ if gene i is “interesting” and $y_i = 0$ otherwise. To perform joint GSEA, we propose to use multiple logistic regression with a sparse *SuSiE* prior on the enrichment parameters, $\boldsymbol{\theta}$:

$$y_i \mid \mathbf{x}_i \sim \text{Bernoulli}(\pi_1(\mathbf{x}_i; \boldsymbol{\theta})); \quad \pi_1(\mathbf{x}_i; \boldsymbol{\theta}) = \frac{\exp \boldsymbol{\theta}^T \mathbf{x}_i}{1 + \exp \boldsymbol{\theta}^T \mathbf{x}_i}; \quad \boldsymbol{\theta} \sim \text{SuSiE} \quad (1)$$

A positive coefficient θ_j corresponds to gene set j being enriched, and a negative coefficient corresponds to being depleted. (We allow for both, although in practice researchers focus on enriched gene sets.) Note: for notational simplicity we omit the intercept term from (1) but will of course include it in our model.

Fitting (1) essentially involves extending our existing *SuSiE* methods to logistic regression. To do this, we will use a standard approach for applying variational methods to logistic regression [124], which iterates between (i) bounding the logistic function by a quadratic function, and (ii) applying variational updates for linear regression (which reduces to regular *SuSiE*).

Related work. Although almost all GSEA analyses are performed marginally, a recent preprint [80] outlines a joint GSEA approach based on (1) with an Elastic Net (EN) penalty [125] instead of our proposed *SuSiE* prior. Both approaches can reduce redundancy because, conditional on one annotation being selected, other correlated annotations will not be selected unless they provide additional enrichment. However, *SuSiE* has several benefits over penalized methods like EN [1]. In particular, *SuSiE* produces CSs that summarize the uncertainty in which GSs are enriched. Each CS is designed to capture an independent enrichment signal, and each CS can contain multiple correlated/redundant GSs that are competing candidates for explaining that signal. In comparison,

EN simply produces a list of potentially-enriched GSs, some of which may be competing explanations of the same signal, with no assessment of uncertainty.

Applications beyond GSEA. Another outcome of this work will be a new variant of *SuSiE*, “logistic *SuSiE*”, that will be generally applicable to variable selection problems with a binary outcome. One such application is fine-mapping of binary disease traits. Currently, binary traits are usually fine-mapped using methods designed for summary data [2]; we will use simulations to assess the benefits of logistic *SuSiE* over these existing approaches.

Aim 2b: Joint GSEA from effect sizes and standard errors. The methods in Aim 2a require only a list of interesting genes. This approach is simple and widely used, but has some disadvantages; a list is usually created by thresholding quantitative values (e.g., effect size estimates or p values), which risks discarding useful information and involves selecting a threshold. To address this, we propose methods that work directly with quantitative assessments. Specifically, following [126], we assume that for each gene i we have an estimated effect size $\hat{\beta}_i$ of some underlying true effect β_i with corresponding standard error s_i . For example, in a differential expression experiment, β_i could be an estimate of the log-fold change in expression of gene i in treatment vs. control. Accounting for different standard errors s_i in different tests helps avoid problems with lower-power tests “polluting” the signal from higher-power tests [126]. However, if effect estimates and standard errors are not available, this framework can also work with z scores and p values; if only a z score is available for each gene, set $\hat{\beta}_i = z_i$ and $s_i = 1$; if only a p value is available, the p values can be translated to z scores.

To adapt these ideas to GSEA, we assume the “two-groups model,” [127],

$$\hat{\beta}_i \mid \beta_i \sim N(\beta_i, s_i^2); \quad \beta_i \mid x_i \sim (1 - \pi_1(x_i; \theta))g_0 + \pi_1(x_i; \theta)g_1; \quad \theta \sim \text{SuSiE} \quad (2)$$

in which g_0 represents the distribution of null effects (usually a point mass at zero), and g_1 represents the distribution of non-null effects, which is to be estimated. As in (1), enrichment is modelled through the effect on π_1 , which is the proportion of non-null effects, and GSs with $\theta_j > 0$ are enriched. To fit (2), we will use an EM algorithm [128] based on the following latent variable representation:

$$\hat{\beta}_i \mid \beta_i \sim N(\beta_i, s_i^2); \quad \beta_i \mid y_i \sim (1 - y_i)g_0 + y_i g_1; \quad y_i \mid x_i \sim \text{Bernoulli}(\pi_1(x_i; \theta)) \quad (3)$$

where $y_i \in \{0, 1\}$. The EM algorithm iterates between computing conditional expectations of β_i and y_i given current parameter estimates (the E-step), and fitting logistic *SuSiE* (1) with the latent terms (y_i) replaced by their conditional expectations (the M-step). Thus, the logistic *SuSiE* methods in Aim 1a become part of an iterative solution to Aim 1b. It is also straightforward to estimate g_1 by maximum-likelihood within this EM framework, and we will implement this for various prior families for g_1 (e.g., normal, and mixture of normals [6]).

Enrichment feedback. A nice feature of this approach is that it allows information on inferred enrichments to inform inference about which genes are differentially expressed [129]. That is, given an estimate of the enrichment parameter, $\hat{\theta}$, the posterior distribution $p(\beta_i \mid \hat{\beta}_i, s_i, y_i, x_i, \hat{\theta})$ can be used to perform inference for β_i , taking account of enrichments. This will have the effect of upweighting tests and improving power for differentially expressed genes that belong to enriched GSs. Incorporating this enrichment information into tests for differential expression is not possible with a one-GS-at-a-time approach to GSEA.

Applications beyond GSEA and related work. Although we will focus initially on GSEA, the model (2) is much more widely applicable to multiple testing applications where many annotations (sometimes called covariates or “side-information”) are available on each test. Such applications are common in genomics. The two-groups model, and other similar models, have been used with a *single* annotation in many genomic applications, including enrichment of GWAS associations [129], and colocalization tests [130]. In the multiple testing literature, various related approaches have been proposed, centered around “covariate-moderated FDR” [131–134]. The key novelty of our approach compared with previous work is our use of the *SuSiE* prior on θ . This will allow our methods to efficiently select among a *large number of correlated annotations*, and summarize the enrichment results in a simple way by reporting CSs of complimentary enrichments.

Extensions. Two limitations of (2) are that: (i) enrichment influences only π_1 , and not the distribution of non-null effects, g_1 ; and (ii) the model is symmetric in positive and negative effects. Once we have an initial version of this method working, we will investigate extensions to address these limitations. This could be done

by extending the two-groups model to a larger number of groups, with different groups capturing positive vs. negative effects, or large vs. small effects (e.g., as in the adaptive shrinkage priors from [6]), and allowing group proportions to depend on the annotations by exploiting the approach in [135].

Benchmarks and comparisons with other approaches. We will assess the performance of our methods in simulated and real data sets. We will simulate data with various enrichment structures (strong vs. weak enrichment, multiple enrichments with varying degrees of overlap, *etc.*) and assess the performance of our methods in comparison to (1) other variations of sparse regression, including EN [80] and (2) popular marginal GSEA methods. In particular, we will compare the *SuSiE* CSs with *post hoc* clustering. We will also compare results for a wide variety of published real data examples where GSEA has been applied.

User-friendly software. To maximize the impact of our work, we will develop an open source R package that allows our approach to be applied to GSEA within an accessible, unified interface. This package will make it straightforward to conduct GSEA against standard gene lists such as those from Gene Ontology [121, 122], Pathway Commons [136, 137], MSigDB [138, 139] and BioSystems [140]. A key advantage of our approach is that, by avoiding redundancy, the output is more concise, and so user-friendly, than marginal analyses.

Potential pitfalls and alternative approaches. Like many existing GSEA methods, our models assume independent tests. In practice, tests are often correlated (e.g., due to gene co-expression, or LD among SNPs), which could cause anti-conservative behavior. We will assess sensitivity of our approach to this issue, and compare to other methods. If necessary, we will investigate allowing for correlations among tests, e.g. using the RSS framework [141], which has already found success for marginal enrichment analysis with correlated tests [141].

Preliminary Results. To illustrate the promise of our proposed approach, we implemented a preliminary version of logistic *SuSiE* (proposed in Aim 2a) and applied it to the CD19+ B cell genes described in “Preliminary Studies.” We allowed up to $L = 10$ CSs (*i.e.*, independent enrichment signals), and *SuSiE* returned 6 CSs; see Table below (p -values are from Fisher’s exact test, FET). Note how the CSs automatically “cluster” similar GSs that are alternative explanations for the same enrichment signal. And by substantially reducing redundancy the results highlight several complementary enrichments that are consistent with known B cell biology, including strong enrichment for an MHC-related GS (CS 5). This enrichment, which is the strongest (largest θ) of all those identified, is swamped in the marginal analysis by redundant results (both GSs rank 90th or lower by FET).

To highlight advantages of *SuSiE* compared with the EN approach of [80], we ran logistic regression with EN regularization (penalty chosen by cross validation; CV). This selected 207 GSs.

CS	gene set	description	genes	θ_j	p value	FET rank
1	GO:0006955	immune response	1,531	1.18	1.41e-47	1
2	GO:0006613	cotranslational protein targeting to membrane	97	3.62	2.47e-31	6
2	GO:0006614	SRP-dependent cotranslational protein targeting to membrane	93	3.62	1.96e-30	8
3	GO:0005615	extracellular space	2,093	0.78	1.16e-29	12
3	GO:0044421	extracellular region part	2,210	0.74	8.25e-28	16
4	GO:0050851	antigen receptor-mediated signaling pathway	169	1.57	1.13e-24	23
5	GO:0023026	MHC class II protein complex binding	15	5.38	1.29e-13	100
5	GO:0023023	MHC protein complex binding	19	4.66	6.74e-14	94
6	GO:0098800	inner mitochondrial membrane protein complex	137	2.41	3.57e-12	128

One explanation for the much larger number of selected GSs is that CV aims to optimize prediction accuracy, whereas *SuSiE* aims to optimize variable selection accuracy. Thus, *SuSiE* provides a shorter list of enrichments that have strong support. Further, *SuSiE* has the advantage that its CSs *automatically organizes* the enriched GSs into those that are potentially redundant (in the same CS) and those that are complementary (in different CSs). In contrast, EN simply lists 207 GSs, with no indication which may be redundant and which complementary.

Aim 3: Develop and apply new methods to discover overlapping cluster structure in genomic data.

Overview. Here we propose flexible new methods to identify structure in genomic data. These methods compliment existing widely used methods such as clustering, principal components analysis, admixture models, topic models and NMF. We use the phrase “overlapping cluster structure” for brevity, but the proposed methods are designed to flexibly capture both discrete and continuous components of variation.

Preliminary Studies: Empirical Bayes Matrix Factorization (EBMF). Our proposed work builds on the EBMF framework developed in the current funding cycle [3]. EBMF factorizes a data matrix Y (an $n \times p$ matrix of n samples measured at p features) into a product of two smaller matrices, $Y = LF' + E$, where the matrices L ($n \times K$) and F ($p \times K$) are to be estimated, and the matrix E ($n \times p$) represents the error in the approximation

$Y \approx LF'$. EBMF introduces prior distributions on F and L that are *learned from the data*. Specifically, EBMF assumes $l_{ik} \sim g_k^l$ and $f_{jk} \sim g_k^f$, and learns the priors $g_1^l, \dots, g_K^l, g_1^f, \dots, g_K^f$ from the data by Variational Empirical Bayes (VEB). The method is tractable for large datasets (e.g., tens of thousands of samples and variables). EBMF also provides an automatic way to select the number of components, K [3, 13]. Rather than being a single method, EBMF is a collection of methods indexed by the prior families chosen for g^l and g^f (which can be different). EBMF uses a modular fitting procedure that makes it easy to implement new families.

Preliminary Studies: Hierarchical structure captured by GBCD. Our proposed work involves two key elements: (i) modify EBMF to factorize YY' instead of Y ; (ii) introduce a generalized binary (GB) prior for L , which encourages L to have entries near 0 or 1 (“quasi-binary”). In this way, we find a factorization $YY' \approx LDL'$ where L is quasi-binary, and D is a diagonal matrix. Because YY' is proportional to the sample covariance matrix (if rows of Y are centered) we call this approach “Generalized Binary Covariance Decomposition” (GBCD).

A key motivation for GBCD is that it can capture hierarchical structures that existing methods like topic models, admixture models, and NMF do not capture. simulated data from four populations A–D related by a simple bifurcating tree (Figure 2, panel a), in which pairs (A, B) and (C, D) are more closely related than other pairs. ADMIXTURE (or NMF or topic models) with $K = 4$ naturally fits these data by using one component for each population (not shown); however, this fails to capture the hierarchical relationships between the populations. Increasing K to 6 simply adds two more non-existent populations (panel b, top). By contrast, a preliminary version of GBCD (see Approach 1 below) with $K = 6$ captures the hierarchical structure by using one component for each branch of the tree (Figure 2, panel b, bottom). In brief, this happens because the (expected) covariance matrix for samples from these four populations is given by

$$\Sigma = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & 2 \end{pmatrix} = LL', \text{ with } L = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

That is, the expected covariance matrix admits a factorization $\Sigma = LL'$ for binary L .

More generally, we can prove that covariance matrices generated by bifurcating trees admit a factorization $\Sigma = LDL'$, where L is a binary matrix with one column for each branch of the tree, and D is a diagonal matrix, with diagonal element d_{kk} capturing the length of branch k . *Note: our methods do not assume a hierarchical structure, nor a strict binary structure, so they can capture other discrete and continuous overlapping structures. This example illustrates that GBCD can capture hierarchical structure when it exists.*

Proposed Work.

Aim 3a: GBCD. We propose to implement GBCD to obtain factorizations of the form $YY' \approx LDL'$, where L is quasi-binary and D is diagonal. For convenience, we rewrite this as $YY' \approx \tilde{L}\tilde{L}'$ where $\tilde{L} = D^{1/2}L$. If L is binary (0 or 1), the k th column of $\tilde{L}$ takes values 0 or $\sqrt{d_{kk}}$, so $\tilde{L}$ is also “binary” (takes two values). For simplicity, we simply use L for $\tilde{L}$ from now on.

In practice, some components of variation may be more continuous than discrete, so to accommodate this we introduce the following *generalized binary* (GB) prior: $p(x | \pi_0, \sigma^2, \mu) = \pi_0 \delta_0(x) + (1 - \pi_0)N(x; \mu, \sigma^2)$. We call this a “generalized binary” prior because as $\sigma^2 \rightarrow 0$, the prior’s support takes only two values: 0 and μ . This GB prior also retains some of the benefits of sparsity and computational convenience of normal distributions. We will use a separate prior for each column of L , with hyperparameters π_0 (which controls sparsity), μ and σ learned from the data. We constrain $\mu > 0$ and require σ^2 to be small, so the GB prior is effectively non-negative. To implement GBCD we will investigate two approaches.

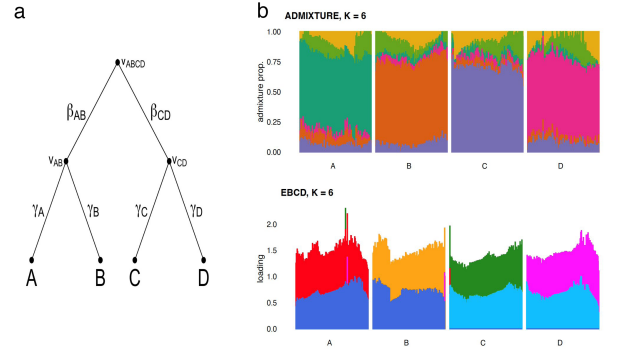


Figure 2: Simulated example with 40 samples from four populations A–D related by a bifurcating tree (a). Results for ADMIXTURE and GBCD shown as stacked bar charts (b). Each sample is shown as a vertical bar, samples ordered by population. The height of each colored bar represents “membership” in each component. Both methods capture the four populations with population-specific components, but only GBCD captures the shared branches of populations (A, B) (dark blue), and of (C, D) (light blue).

Approach 1: Apply the existing EBMF algorithm to factorize YY' instead of Y with GB prior on both L and F . This yields factorization $YY' \approx LF'$ with GB prior on L and F . Because the input matrix YY' is symmetric, and the priors on L and F are the same, estimated values of L and F should be similar, and our preliminary investigations confirm this. This approach therefore yields a factorization $YY' \approx LF' \approx LL'$ with GB prior on L .

Approach 2: Modify EBMF to constrain F to be orthonormal, $F'F = I_p$, where I_p is the $p \times p$ identity matrix, and apply this modified EBMF to the data matrix Y with GB prior on L . This approach will yield a factorization $Y \approx LF'$ where $F'F = I_p$, which implies $YY' \approx LF'FL' = LL'$. To incorporate the constraint $F'F = I_p$ into EBMF, we make use of the following result [142]: under the EBMF model, the maximum-likelihood estimate of F subject to $F'F = I_p$ is $F = \text{Polar}(Y'L)$, where we define $\text{Polar}(X) = UV'$ where $X = UDV'$ is the singular value decomposition of matrix X . This constrained update for F can be inserted into the EBMF algorithm of [3] to replace its usual (unconstrained) updates.

Approach 1 has the advantage that it requires very little additional implementation; thanks to the modularity of the EBMF fitting algorithm [3], it requires only implementing a solver for the Empirical Bayes Normal Means problem [13] with a GB prior, which can be done using numerical optimization. We therefore implemented this approach to generate Preliminary Results. However, Approach 2 has some advantages; it is more principled, and allows for a joint update of all factors simultaneously, which may improve convergence compared with EBMF which must update one factor at a time. Approach 2 also avoids forming the matrix YY' , which in some applications could be large. We will therefore develop both approaches and compare them in simulated and real data sets. Since GBCD is a non-convex optimization problem, results may depend on initialization (the same is true of ADMIXTURE, topic models and NMF), and we will investigate several initialization approaches.

Estimating F . Whichever approach is used, once we obtain L such that $YY' \approx LL'$, we will also estimate a corresponding F such that $Y \approx LF'$. This can be useful because F captures how the different variables contribute to the components of variation. For example, in single-cell applications, if L captures overlapping cluster structure in the cells, F captures which genes contribute to each component, which could be critical to interpretation. Approach 2 automatically provides an estimate of F ; for Approach 1, we will take the estimated L and apply EBMF (with L fixed) to find F .

Benchmarks. We will perform extensive simulations to test our methods and compare them with one another and with existing methods (including topic models, admixture models, NMF, hierarchical clustering, clustering, and PCA). To aid understanding, we will simulate under toy scenarios (e.g. simple hierarchical scenarios where hierarchical clustering should do well; simple admixture scenarios where admixture should do well) as well as more complex scenarios involving mixed discrete and continuous components. We emphasize that we see a continuing role for all the competing analysis methods; a major goal of the simulations will be to assess when our new methods provide different representations than existing methods, and therefore perhaps new insights.

Aim 3b: Data applications. To assess our methods, we will apply them to a wide range of genomic applications:

Population genetics. Our preliminary results suggest that our proposed methods may be particularly well-suited to identifying shared components of ancestry from population genetic data. We will investigate this in large diverse samples, e.g. the Human Origins array genotype data from 147 worldwide populations [143]. An intriguing possibility is that our methods will help provide new insights into analyses involving ancient DNA (e.g., [143–145]) by highlighting components of ancestry that are shared between modern and ancient samples. We will assess this possibility and compare results with published results and results from other tools specifically designed for these applications, including ADMIXTURE [82] and qpgraph [146].

Single-cell tumor data. Tumor data raise particular challenges because cells tend to cluster by patient (due to strong patient-specific effects) which can obscure other interesting structure, e.g. patients that share a disease subtype [147, 148]. GBCD should be particularly helpful for capturing this overlapping structure, as well as structure due to dataset-specific effects ("batch effects"). To assess this, we will analyze data from several studies on pancreatic ductal adenocarcinoma (PDAC) [149–152]. The heterogeneity across studies (e.g., some are primary tumors, others are metastatic) combined with patient-specific effects make this particularly challenging.

For these analyses we must decide whether and how to transform the observed count matrix (a sparse matrix containing read or UMI counts) to produce the data matrix Y for analysis. Applications of EBMF to single-cell

data in [13] suggest a log-transform, with a “pseudocount” to deal with zero can work surprisingly well, but we will also explore other approaches, including forming Pearson residuals [153], and bi-whitening [154]. We will compare results with other methods used to discover structure in single-cell data, including *t*-SNE [155], UMAP [156], topic models [11, 12], and glmpca [153].

Genomic data where NMF have been applied. Our methods provide a natural alternative to NMF: whereas NMF seeks a non-negative factorization of the data matrix Y , GBCD seeks a non-negative (quasi-binary) factorization of YY' . As our Preliminary results demonstrate, this can produce surprisingly different results. Therefore we will compare GBCD with NMF on other genomics applications where NMF has been applied, including analyses of mutational signatures for somatic mutations [92, 93] and for germ line mutations [94].

We will use results from all applications to identify limitations of our methods, and, where possible, develop ways to address these limitations and improve the methods. We will also use these applications to develop approaches to help users *interpret* the components identified, e.g., in single-cell data to identify genes that contribute to each component, and to visualize results (e.g. expanding on barplots and heatmaps below).

Incorporating covariates.. Sample-specific covariates can be accounted for by including them as “fixed” (rather than estimated) columns of L . We will implement this and assess how incorporating covariates (e.g. sex, cohort, patient) affects results in our data analyses.

Preliminary Results. To illustrate the potential of GBCD we implemented **Approach 1**, and applied it to 2,504 genotype samples from Phase 3 of the 1000 Genomes Project [157]. For comparison, we applied ADMIXTURE [82], the most widely used approach for such data. ADMIXTURE separates major geographic groups—Africa, Europe, India, and East Asia—with little to no sharing of ancestry components between groups (Figure 3, top). In particular, three sample populations—LWK from Kenya, FIN from Finland and JPT from Japan—are each largely captured by their own component not appearing elsewhere. It does capture European admixture in African Americans (ASW) and in the Americas (CLM, MXL, PUR, PEL). By contrast, the GBCD results emphasize shared ancestry among many groups; for example, component k1 captures sharing across Europe and Asia, presumably representing an “out of Africa” feature. And, in contrast to the ADMIXTURE results, LWK, FIN and JPT show considerable sharing with other related groups: LWK samples share a large component with the rest of Africa, JPT samples share many components with China, and FIN samples shares several (but not all) components with other European groups, as well as one (k7) with Asians, highlighting a historic connection between Finns and Asians [158]. The populations traditionally described as “admixed” are each captured by a mix of components shared with Africa, Europe and Asia. While much of the structure is near-binary and hierarchical, not all of it is; for example, k7, k9 and k11 overlap but not in a hierarchical way, because human history does not conform to a strict bifurcating pattern.

Single-cell RNA-seq data from primary Pancreatic ductal adenocarcinoma (PDAC) patients. We analyzed data from three recent studies reporting scRNA-seq from PDAC patients [149, 150, 152]. We analyzed malignant cells from (histology-confirmed) tumor samples. After quality control, the data contain UMI counts for 13,844 genes measured in 35,670 malignant cells from 37 primary and 22 metastatic tumors from both sexes. *t*-SNE dimension reduction [159] highlights strong cluster structure: cells cluster by patient due to substantial inter-patient tumor heterogeneity and copy number variation [149]. This feature, typical of tumor scRNA-seq data, makes it hard to identify shared patterns, e.g. cancer subtypes, among patients [147, 148].

We applied **Approach 1** to estimate quasi-binary L , then estimate F by applying EBMF to obtain $Y \approx LF'$. L_{ik} represents membership of cell i in factor k and F_{jk} represents change in expression of gene j due to membership

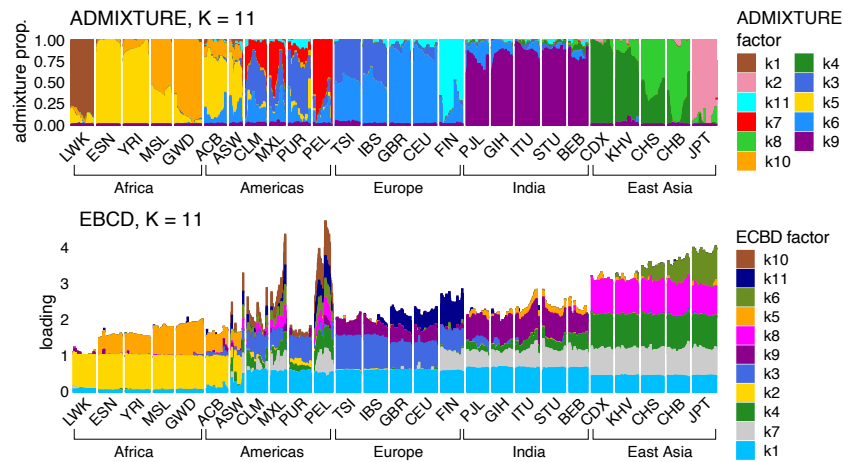


Figure 3: Comparison of ADMIXTURE [82] (top) and a preliminary implementation of GBCD (bottom) applied to genotype data from Phase 3 of 1000 Genomes project (2,504 individuals, 184,224 SNPs) [157].

in factor k . Thus each factor represents a change in expression of some genes in some cells, so we call the factors “gene expression programs” (GEPs). To interpret each GEP we examine which genes contribute most strongly (largest $|F_{jk}|$) and perform GSEA on these “key” genes.

Consistent with known strong patient-specific effects, GBCD identified many near-binary GEPs that appear in only one or a few patients (GEP17-34), as well as cohort specific factors (“batch” effects; GEPs15-16). However, GBCD also identifies factors (GEPs 1-14) that are shared across patients. This highlights a key feature of GBCD: *strong patient (and/or batch) effects do not preclude identifying other biological factors that are shared across patients*. Analyses of key genes show that GEPs 1-8 correlate with clinically relevant PDAC cancer subtypes (classical and basal) previously identified based on gene expression patterns [150, 152, 160–162]. These 8 GEPs are quite distinct, suggesting considerable structure beyond a simple classical/basal dichotomy. In contrast, GEPs 9-14 are unassociated with previously-identified subtype signatures. GEP10 is enriched for cell cycle genes, and likely reflects the cycle status of each cell. Other GEPs are enriched for interferon signaling (GEP9), TNF-NF κ B signaling (GEP11), respiratory electron transport (GEP12), translation (GEP13) and stress (GEP14).

Existing scRNA-seq datasets are limited in size (number of tumors) and clinical characterization. Therefore, to assess potential clinical relevance of GEPs 1-14 we scored the GEPs, as well as previously-published classical/basal signatures, in a larger bulk RNA-seq dataset of 391 resected primary PDAC tumors from 4 studies [163–166] which include data on age, sex, tumor stage and right-censored time of death (260 deaths). Stepwise regression (Cox proportional hazards model with survival as endpoint, adjusting for age, sex, and tumor stage) selected two of our GEPs as associated with shorter survival: GEP4 (Hazard Ratio, HR = 1.58, P -value = $2e-10$) is the only selected variable related to classical/basal subtype, indicating that it has stronger association with patient prognosis than previously defined subtype signatures; and GEP14 (HR = 1.32, P -value = $8e-5$), which represents a potentially novel gene signature predictive of poor prognosis independently of classical/basal subtype and tumor stage. GEP14 is enriched for genes up-regulated in response to hypoxia, and pancreatic cancer is commonly characterized by severely hypoxic microenvironment, which has been shown to correlate with tumor progression and poor survival, and has been a promising target of innovative therapeutic strategies [167, 168]. We have presented and discussed these preliminary results with the Pancreas Working Group at UChicago, who will continue to advise us as our methods and results mature to publication (see letter of support from Professors K. Macleod and S. Oakes).

Our preliminary results show the potential of GBCD to capture complex overlapping and hierarchical structure, both discrete (e.g. patient effects) and continuous (e.g. cell cycle), that existing methods struggle to capture.

Timeline. Work on Aims 1–3 will proceed in parallel with following milestones: **Year 1:** Complete initial implementation and testing of *fSuSiE* (Aim 1a), GSEA methods (Aim 2a), and GBCD (Aim 3a). **Year 2:** Submit manuscripts and publicly distribute fully documented software for Aims 1a (with applications to data, 1c) and 2a. Complete population genetic applications of GBCD (Aim 3b). **Year 3:** Submit manuscript and distribute fully documented software for GBCD (Aims 3a, 3b). Complete initial investigation of other GBCD applications (Aim 3b). Have working version of *fSuSiE* for multiple traits (Aim 1b) and GSEA for quantitative effects (Aim 2b). **Year 4:** Submit second manuscript on each Aim (1b, 1c, 2b, 3b), with accompanying distributable software.

Summary. *This project will create methods and software for several important genomics applications, applicable to data from both sexes and all ancestries. It will make these methods conveniently available in software for the broad community of genome scientists who use these methods in their research so that all may benefit from this work.*

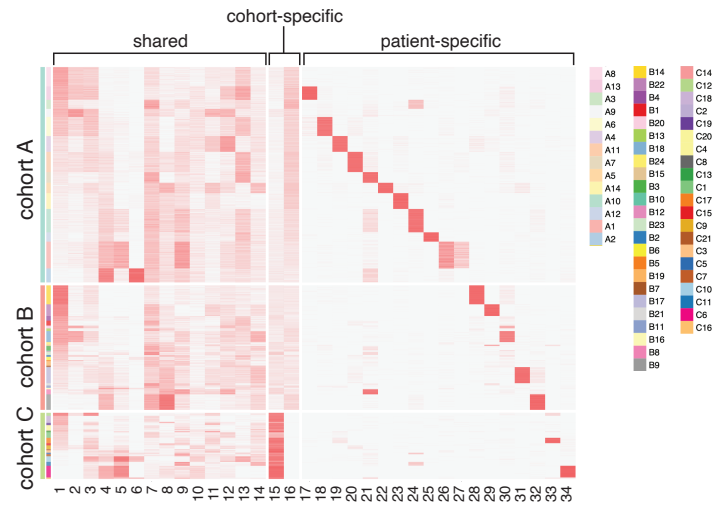


Figure 4: Preliminary GBCD results on PDAC data. Heatmap shows memberships (L) of 35,670 cells (rows) in 34 GEPs (columns), on a scale from 0 (white) to 1 (dark red). Cells are ordered by cohort and patient (color bars, left).

Human Subjects Research

No; we will analyze existing and anonymized data only.

Inclusion of Women.

Methods developed will be applicable to both sexes.

Inclusion of Minorities.

Methods developed will be applicable to all ethnicities.

Inclusion of Children.

Methods developed will be applicable to all ages.

Literature Cited

- [1] Wang, G., Sarkar, A., Carbonetto, P., and Stephens, M.: A simple new approach to variable selection in regression, with application to genetic fine mapping. Journal of the Royal Statistical Society, Series B. 82(5): 1273–1300, 2020.
- [2] Zou, Y., Carbonetto, P., Wang, G., and Stephens, M.: Fine-mapping from summary data with the “sum of single effects” model. PLoS Genetics. 18(7): e1010299, 2022.
- [3] Wang, W. and Stephens, M.: Empirical Bayes matrix factorization. Journal of Machine Learning Research. 22(120): 1–40, 2021.
- [4] Servin, B. and Stephens, M.: Imputation-based analysis of association studies: candidate regions and quantitative traits. PLoS Genetics. 3(7): e114, 2007.
- [5] Veyrieras, J. B., Kudaravalli, S., Kim, S. Y., Dermitzakis, E. T., Gilad, Y., Stephens, M., and Pritchard, J. K.: High-resolution mapping of expression-QTLs yields insight into human gene regulation. PLoS Genetics. 4(10): e1000214, 2008.
- [6] Stephens, M.: False discovery rates: a new deal. Biostatistics. 18(2): 275–294, 2017.
- [7] Urbut, S. M., Wang, G., Carbonetto, P., and Stephens, M.: Flexible statistical methods for estimating and testing effects in genomic studies with multiple conditions. Nature Genetics. 51(1): 187–195, 2019.
- [8] Pritchard, J. K., Stephens, M., and Donnelly, P.: Inference of population structure using multilocus genotype data. Genetics. 155(2): 945–959, 2000.
- [9] Novembre, J., Johnson, T., Bryc, K., Kutalik, Z., Boyko, A. R., Auton, A., Indap, A., King, K. S., Bergmann, S., Nelson, M. R., Stephens, M., and Bustamante, C. D.: Genes mirror geography within Europe. Nature. 456(7218): 98–101, 2008.
- [10] Engelhardt, B. E. and Stephens, M.: Analysis of population structure: a unifying framework and novel methods based on sparse factor analysis. PLoS Genetics. 6(9): e1001117, 2010.
- [11] Dey, K. K., Hsiao, C. J., and Stephens, M.: Visualizing the structure of RNA-seq expression data using grade of membership models. PLoS Genetics. 13(3): e1006599, 2017.
- [12] Carbonetto, P., Sarkar, A., Wang, Z., and Stephens, M.: Non-negative matrix factorization algorithms greatly improve topic model fits. arXiv. 2105.13440, 2021.
- [13] Willwerscheid, J. and Stephens, M.: ebnm: an R package for solving the empirical Bayes normal means problem using a variety of prior families. arXiv. 2110.00152, 2021.
- [14] Hutchinson, A., Asimit, J., and Wallace, C.: Fine-mapping genetic associations. Human Molecular Genetics. 29(R1): R81–R88, 2020.
- [15] Maller, J. B., McVean, G., Byrnes, J., Vukcevic, D., Palin, K., Su, Z., Howson, J. M. M., Auton, A., Myers, S., Morris, A., Pirinen, M., Brown, M. A., Burton, P. R., Caulfield, M. J., Compston, A., Farrall, M., Hall, A. S., Hattersley, A. T., Hill, A. V. S., Mathew, C. G., Pembrey, M., Satsangi, J., Stratton, M. R., Worthington, J., Craddock, N., Hurles, M., Ouwehand, W., Parkes, M., Rahman, N., Duncanson, A., Todd, J. A., Kwiatkowski, D. P., Samani, N. J., Gough, S. C. L., McCarthy, M. I., Deloukas, P., Donnelly, P., and Consortium, T. W. T. C. C.: Bayesian refinement of association signals for 14 loci in 3 common diseases. Nature Genetics. 44(12): 1294–1301, 2012.
- [16] Schaid, D. J., Chen, W., and Larson, N. B.: From genome-wide associations to candidate causal variants by statistical fine-mapping. Nature Reviews Genetics. 19(8): 491–504, 2018.

- [17] Giambartolomei, C., Vukcevic, D., Schadt, E. E., Franke, L., Hingorani, A. D., Wallace, C., and Plagnol, V.: Bayesian test for colocalisation between pairs of genetic association studies using summary statistics. PLoS Genetics. 10(5): e1004383, 5 2014.
- [18] Wallace, C.: Statistical testing of shared genetic control for potentially related traits. Genetic Epidemiology. 37(8): 802–813, 2013.
- [19] Wallace, C.: Eliciting priors and relaxing the single causal variant assumption in colocalisation analyses. PLoS Genetics. 16(4): e1008720, 2020.
- [20] Wallace, C.: A more accurate method for colocalisation analysis allowing for multiple causal variants. PLoS Genetics. 17(9): e1009440, 2021.
- [21] Foley, C. N., Staley, J. R., Breen, P. G., Sun, B. B., Kirk, P. D. W., Burgess, S., and Howson, J. M. M.: A fast and efficient colocalization algorithm for identifying shared genetic risk factors across multiple traits. Nature Communications. 12: 764, 2021.
- [22] Pickrell, J. K., Marioni, J. C., Pai, A. A., Degner, J. F., Engelhardt, B. E., Nkadori, E., Veyrieras, J.-B., Stephens, M., Gilad, Y., and Pritchard, J. K.: Understanding mechanisms underlying human gene expression variation with RNA sequencing. Nature. 464(7289): 768–772, 2010.
- [23] Degner, J. F., Marioni, J. C., Pai, A. A., Pickrell, J. K., Nkadori, E., Gilad, Y., and Pritchard, J. K.: Effect of read-mapping biases on detecting allele-specific expression from RNA-sequencing data. Bioinformatics. 25(24): 3207–3212, 2009.
- [24] Bock, C.: Analysing and interpreting DNA methylation data. Nature Reviews Genetics. 13(10): 705–719, 2012.
- [25] Teschendorff, A. E. and Relton, C. L.: Statistical and integrative system-level analysis of DNA methylation data. Nature Reviews Genetics. 19(3): 129–147, 2018.
- [26] Spitz, F. and Furlong, E. E. M.: Transcription factors: from enhancer binding to developmental control. Nature Reviews Genetics. 13(9): 613–626, 2012.
- [27] Yan, J., Qiu, Y., Ribeiro dos Santos, A. M., Yin, Y., Li, Y. E., Vinckier, N., Nariyai, N., Benaglio, P., Raman, A., Li, X., Fan, S., Chiou, J., Chen, F., Frazer, K. A., Gaulton, K. J., Sander, M., Taipale, J., and Ren, B.: Systematic analysis of binding of transcription factors to noncoding variants. Nature. 591(7848): 147–151, 2021.
- [28] Zhou, V. W., Goren, A., and Bernstein, B. E.: Charting histone modifications and the functional organization of mammalian genomes. Nature Reviews Genetics. 12(1): 7–18, 2011.
- [29] Morgan, M. A. J. and Shilatifard, A.: Reevaluating the roles of histone-modifying enzymes and their associated chromatin modifications in transcriptional regulation. Nature Genetics. 52(12): 1271–1281, 2020.
- [30] Benner, C., Spencer, C. C. A., Havulinna, A. S., Salomaa, V., Ripatti, S., and Pirinen, M.: FINEMAP: efficient variable selection using summary data from genome-wide association studies. Bioinformatics. 32(10): 1493–1501, 2016.
- [31] Benner, C., Havulinna, A. S., Järvelin, M.-R., Salomaa, V., Ripatti, S., and Pirinen, M.: Prospects of fine-mapping trait-associated genomic regions by using summary statistics from genome-wide association studies. American Journal of Human Genetics. 101(4): 539–551, 10 2017.
- [32] Benner, C., Havulinna, A. S., Salomaa, V., Ripatti, S., and Pirinen, M.: Refining fine-mapping: effect sizes and regional heritability. bioRxiv. doi:10.1101/318618, 2018.

- [33] Chen, W., Larrabee, B. R., Ovsyannikova, I. G., Kennedy, R. B., Haralambieva, I. H., Poland, G. A., and Schaid, D. J.: Fine mapping causal variants with an approximate Bayesian method using marginal test statistics. Genetics. 200(3): 719–736, 2015.
- [34] Hormozdiari, F., Kostem, E., Kang, E. Y., Pasaniuc, B., and Eskin, E.: Identifying causal variants at loci with multiple signals of association. Genetics. 198(2): 497–508, 2014.
- [35] Kichaev, G., Yang, W.-Y., Lindstrom, S., Hormozdiari, F., Eskin, E., Price, A. L., Kraft, P., and Pasaniuc, B.: Integrating functional data to prioritize causal variants in statistical fine-mapping studies. PLoS Genetics. 10(10): e1004722, 2014.
- [36] Lee, Y., Luca, F., Pique-Regi, R., and Wen, X.: Bayesian multi-SNP genetic association analysis: control of FDR and use of summary statistics. bioRxiv. doi:10.1101/316471, 2018.
- [37] Newcombe, P. J., Conti, D. V., and Richardson, S.: JAM: a scalable Bayesian framework for joint analysis of marginal SNP effects. Genetic Epidemiology. 40(3): 188–201, 2016.
- [38] Wallace, C., Cutler, A. J., Pontikos, N., Pekalski, M. L., Burren, O. S., Cooper, J. D., García, A. R., Ferreira, R. C., Guo, H., Walker, N. M., Smyth, D. J., Rich, S. S., Onengut-Gumuscu, S., Sawcer, S. J., Ban, M., Richardson, S., Todd, J. A., and Wicker, L. S.: Dissection of a complex disease susceptibility region using a Bayesian stochastic search approach to fine mapping. PLoS Genetics. 11(6): e1005272, 2015.
- [39] Wen, X., Lee, Y., Luca, F., and Pique-Regi, R.: Efficient integrative multi-SNP association analysis via deterministic approximation of posteriors. American Journal of Human Genetics. 98(6): 1114–1129, 2016.
- [40] Kichaev, G., Roytman, M., Johnson, R., Eskin, E., Lindström, S., Kraft, P., and Pasaniuc, B.: Improved methods for multi-trait fine mapping of pleiotropic risk loci. Bioinformatics. 33(2): 248–255, 2017.
- [41] Lewin, A., Saadi, H., Peters, J. E., Moreno-Moral, A., Lee, J. C., Smith, K. G., Petretto, E., Bottolo, L., and Richardson, S.: MT-HESS: an efficient Bayesian approach for simultaneous association detection in OMICS datasets, with application to eQTL mapping in multiple tissues. Bioinformatics. 32(4): 523–532, 2016.
- [42] Zou, Y. Bayesian variable selection from summary data, with application to joint fine-mapping of multiple traits. PhD thesis, University of Chicago, 2021.
- [43] Robertson, G., Hirst, M., Bainbridge, M., Bilenky, M., Zhao, Y., Zeng, T., Euskirchen, G., Bernier, B., Varhol, R., Delaney, A., and others, : Genome-wide profiles of STAT1 DNA association using chromatin immunoprecipitation and massively parallel sequencing. Nature Methods. 4(8): 651–657, 2007.
- [44] Buenrostro, J. D., Wu, B., Chang, H. Y., and Greenleaf, W. J.: ATAC-seq: a method for assaying chromatin accessibility genome-wide. Current Protocols in Molecular Biology. 109(1): 21.29.1–21.29.9, 2015.
- [45] Lister, R., Pelizzola, M., Dowen, R. H., Hawkins, R. D., Hon, G., Tonti-Filippini, J., Nery, J. R., Lee, L., Ye, Z., Ngo, Q.-M., and others, : Human DNA methylomes at base resolution show widespread epigenomic differences. Nature. 462(7271): 315–322, 2009.
- [46] Hansen, K. D., Langmead, B., and Irizarry, R. A.: BSmooth: from whole genome bisulfite sequencing reads to differentially methylated regions. Genome Biology. 13(10): R83, 2012.
- [47] Shim, H., Chasman, D. I., Smith, J. D., Mora, S., Ridker, P. M., Nickerson, D. A., Krauss, R. M., and Stephens, M.: A multivariate genome-wide association analysis of 10 LDL subfractions, and their response to statin treatment, in 1868 Caucasians. PLoS ONE. 10(4): e0120758, 2015.
- [48] Shim, H., Xing, Z., Pantaleo, E., Luca, F., Pique-Regi, R., and Stephens, M.: Multi-scale Poisson process approaches for differential expression analysis of high-throughput sequencing data. arXiv. 2106.13634, 2021.

- [49] Peters, T. J., Buckley, M. J., Statham, A. L., Pidsley, R., Samaras, K., V Lord, R., Clark, S. J., and Molloy, P. L.: De novo identification of differentially methylated regions in the human genome. Epigenetics & Chromatin. 8: 6, 2015.
- [50] Lee, W. and Morris, J. S.: Identification of differentially methylated loci using wavelet-based functional mixed models. Bioinformatics. 32(5): 664–672, March 2016.
- [51] Subramanian, A., Tamayo, P., Mootha, V. K., Mukherjee, S., Ebert, B. L., Gillette, M. A., Paulovich, A., Pomeroy, S. L., Golub, T. R., Lander, E. S., and Mesirov, J. P.: Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. Proceedings of the National Academy of Sciences. 102(43): 15545–50, 2005.
- [52] Goeman, J. J., van de Geer, S. A., de Kort, F., and van Houwelingen, H. C.: A global test for groups of genes: Testing association with a clinical outcome. Bioinformatics. 20(1): 93–99, 2004.
- [53] Kim, S.-Y. and Volsky, D. J.: PAGE: Parametric analysis of gene set enrichment. BMC Bioinformatics. 6: 144, 2005.
- [54] Barry, W. T., Nobel, A. B., and Wright, F. A.: Significance analysis of functional categories in gene expression studies: A structured permutation approach. Bioinformatics. 21(9): 1943–1949, 2005.
- [55] Tian, L., Greenberg, S. A., Kong, S. W., Altschuler, J., Kohane, I. S., and Park, P. J.: Discovering statistically significant pathways in expression profiling studies. Proceedings of the National Academy of Sciences. 102(38): 13544–13549, 2005.
- [56] Falcon, S. and Gentleman, R.: Using GOstats to test gene lists for GO term association. Bioinformatics. 23(2): 257–258, 2007.
- [57] Lu, Y., Rosenfeld, R., Simon, I., Nau, G. J., and Bar-Joseph, Z.: A probabilistic generative model for GO enrichment analysis. Nucleic Acids Research. 36(17): e109, 2008.
- [58] Sartor, M. A., Leikauf, G. D., and Medvedovic, M.: LRpath: a logistic regression approach for identifying enriched biological groups in gene expression data. Bioinformatics. 25(2): 211–217, 2009.
- [59] Eden, E., Navon, R., Steinfeld, I., Lipson, D., and Yakhini, Z.: GOrilla: a tool for discovery and visualization of enriched GO terms in ranked gene lists. BMC Bioinformatics. 10: 48, 2009.
- [60] Irizarry, R. A., Wang, C., Zhou, Y., and Speed, T. P.: Gene set enrichment analysis made simple. Statistical Methods in Medical Research. 18(6): 565–575, 2019.
- [61] Bauer, S., Gagneur, J., and Robinson, P. N.: GOing Bayesian: model-based gene set analysis of genome-scale data. Nucleic Acids Research. 38(11): 3523–3532, 2020.
- [62] Mi, G., Di, Y., Emerson, S., Cumbie, J. S., and Chang, J. H.: Length bias correction in Gene Ontology enrichment analysis using logistic regression. PLoS ONE. 7(10): e46128, 2012.
- [63] Wu, D. and Smyth, G. K.: Camera: A competitive gene set test accounting for inter-gene correlation. Nucleic Acids Research. 40(17): e133, 2012.
- [64] Geistlinger, L., Csaba, G., and Zimmer, R.: Bioconductor’s EnrichmentBrowser: seamless navigation through combined results of set- & network-based enrichment analysis. BMC Bioinformatics. 17: 45, 2016.
- [65] Dong, X., Hao, Y., Wang, X., and Tian, W.: LEGO: A novel method for gene set over-representation analysis by incorporating network-based gene weights. Scientific Reports. 6: 18871, 2016.
- [66] Simillion, C., Liechti, R., Lischer, H. E., Ioannidis, V., and Bruggmann, R.: Avoiding the pitfalls of gene set enrichment analysis with SetRank. BMC Bioinformatics. 18: 151, 2017.

- [67] Harrison, P. F., Pattison, A. D., Powell, D. R., and Beilharz, T. H.: Topconfects: a package for confident effect sizes in differential expression analysis provides a more biologically useful ranked gene list. Genome Biology. 20: 67, 2019.
- [68] Ma, Y., Sun, S., Shang, X., Keller, E. T., Chen, M., and Zhou, X.: Integrative differential expression and gene set enrichment analysis using summary statistics for scRNA-seq studies. Nature Communications. 11: 1585, 2020.
- [69] Sherman, B. T., Hao, M., Qiu, J., Jiao, X., Baseler, M. W., Lane, H. C., Imamichi, T., and Chang, W.: DAVID: a web server for functional enrichment analysis and functional annotation of gene lists (2021 update). Nucleic Acids Research. gkac194, 2022.
- [70] Zhou, Y., Zhou, B., Pache, L., Chang, M., Khodabakhshi, A. H., Tanaseichuk, O., Benner, C., and Chanda, S. K.: Metascape provides a biologist-oriented resource for the analysis of systems-level datasets. Nature Communications. 10: 1523, 2019.
- [71] Liao, Y., Wang, J., Jaehnig, E. J., Shi, Z., and Zhang, B.: WebGestalt 2019: gene set analysis toolkit with revamped UIs and APIs. Nucleic Acids Research. 47(W1): W199–W205, 2019.
- [72] Raudvere, U., Kolberg, L., Kuzmin, I., Arak, T., Adler, P., Peterson, H., and Vilo, J.: G:Profiler: a web server for functional enrichment analysis and conversions of gene lists (2019 update). Nucleic Acids Research. 47(W1): W191–W198, 2019.
- [73] Kuleshov, M. V., Jones, M. R., Rouillard, A. D., Fernandez, N. F., Duan, Q., Wang, Z., Koplev, S., Jenkins, S. L., Jagodnik, K. M., Lachmann, A., McDermott, M. G., Monteiro, C. D., Gundersen, G. W., and Ma'ayan, A.: Enrichr: a comprehensive gene set enrichment analysis web server 2016 update. Nucleic Acids Research. 44(W1): W90–W97, 2016.
- [74] Tabas-Madrid, D., Nogales-Cadenas, R., and Pascual-Montano, A.: GeneCodis3: a non-redundant and modular enrichment analysis tool for functional genomics. Nucleic Acids Research. 40(W1): W478–W483, 2012.
- [75] Huang, D. W., Sherman, B. T., and Lempicki, R. A.: Bioinformatics enrichment tools: paths toward the comprehensive functional analysis of large gene lists. Nucleic Acids Research. 37(1): 1–13, 2008.
- [76] Reimand, J., Isserlin, R., Voisin, V., Kucera, M., Tannus-Lopes, C., Rostamianfar, A., Wadi, L., Meyer, M., Wong, J., Xu, C., Merico, D., and Bader, G. D.: Pathway enrichment analysis and visualization of omics data using g:Profiler, GSEA, Cytoscape and EnrichmentMap. Nature Protocols. 14(2): 482–517, 2019.
- [77] Fontanillo, C., Nogales-Cadenas, R., Pascual-Montano, A., and Rivas, J. D. L.: Functional analysis beyond enrichment: Non-redundant reciprocal linkage of genes and biological terms. PLoS ONE. 6(9): e24289, 2011.
- [78] Vivar, J. C., Pemu, P., McPherson, R., and Ghosh, S.: Redundancy control in pathway databases (ReCiPa): an application for improving gene-set enrichment analysis in omics studies and "big data" biology. Omics. 17(8): 414–422, 2013.
- [79] Ebrahimipoor, M., Spitali, P., Hettne, K., Tsonaka, R., and Goeman, J.: Simultaneous enrichment analysis of all possible gene-sets: unifying self-contained and competitive methods. Briefings in Bioinformatics. 21(4): 1302–1312, 2019.
- [80] Fang, T., Davydov, I., Marbach, D., and Zhang, J. D.: Gene-set enrichment with regularized regression. bioRxiv. doi:10.1101/659920, 2019.
- [81] Rosenberg, N. A.: Genetic structure of human populations. Science. 298(5602): 2381–2385, 2002.

- [82] Alexander, D. H., Novembre, J., and Lange, K.: Fast model-based estimation of ancestry in unrelated individuals. Genome Research. 19(9): 1655–1664, 2009.
- [83] Blei, D. M., Ng, A. Y., and Jordan, M. I.: Latent Dirichlet allocation. Journal of Machine Learning Research. 3: 993–1022, 2003.
- [84] González-Blas, C., Minnoye, L., Papasokrati, D., Aibar, S., Hulselmans, G., Christiaens, V., Davie, K., Wouters, J., and Aerts, S.: cisTopic: cis-regulatory topic modeling on single-cell ATAC-seq data. Nature Methods. 16(5): 397–400, 2019.
- [85] Bielecki, P., Riesenfeld, S. J., Hütter, J.-C., Torlai Triglia, E., Kowalczyk, M. S., Ricardo-Gonzalez, R. R., Lian, M., Amezcua Vesely, M. C., Kroehling, L., Xu, H., Slyper, M., Muus, C., Ludwig, L. S., Christian, E., Tao, L., Kedaigle, A. J., Steach, H. R., York, A. G., Skadow, M. H., Yaghoubi, P., Dionne, D., Jarret, A., McGee, H. M., Porter, C. B. M., Licona-Limón, P., Bailis, W., Jackson, R., Gagliani, N., Gasteiger, G., Locksley, R. M., Regev, A., and Flavell, R. A.: Skin-resident innate lymphoid cells converge on a pathogenic effector state. Nature. 592: 128–132, 2021.
- [86] Housman, G., Briscoe, E., and Gilad, Y.: Evolutionary insights into primate skeletal gene regulation using a comparative cell culture model. PLoS Genetics. 18(3): e1010073, 2022.
- [87] Hung, A., Housman, G., Briscoe, E. A., Cuevas, C., and Gilad, Y.: Characterizing gene expression in an in vitro biomechanical strain model of joint health. F1000Research. 11: 296, 2022.
- [88] Rhodes, K., Barr, K. A., Popp, J. M., Strober, B. J., Battle, A., and Gilad, Y.: Human embryoid bodies as a novel system for genomic studies of functionally diverse cell types. eLife. 11: e71361, 2022.
- [89] Schenkel, J. M., Herbst, R. H., Canner, D., Li, A., Hillman, M., Shanahan, S.-L., Gibbons, G., Smith, O. C., Kim, J. Y., Westcott, P., Hwang, W. L., Freed-Pastor, W. A., Eng, G., Cuoco, M. S., Rogers, P., Park, J. K., Burger, M. L., Rozenblatt-Rosen, O., Cong, L., Pauken, K. E., Regev, A., and Jacks, T.: Conventional type I dendritic cells maintain a reservoir of proliferative tumor-antigen specific TCF-1+ CD8+ T cells in tumor-draining lymph nodes. Immunity. 54(10): 2338–2353.e6, oct 2021.
- [90] Xu, H., Ding, J., Porter, C. B., Wallrapp, A., Tabaka, M., Ma, S., Fu, S., Guo, X., Riesenfeld, S. J., Su, C., Dionne, D., Nguyen, L. T., Lefkovith, A., Ashenberg, O., Burkett, P. R., Shi, H. N., Rozenblatt-Rosen, O., Graham, D. B., Kuchroo, V. K., Regev, A., and Xavier, R. J.: Transcriptional atlas of intestinal immune cells reveals that neuropeptide α -CGRP modulates group 2 innate lymphoid cell responses. Immunity. 51(4): 696–708, 2019.
- [91] Lee, D. D. and Seung, H. S.: Learning the parts of objects by non-negative matrix factorization. Nature. 401(6755): 788–791, 1999.
- [92] Alexandrov, L. B., Kim, J., Haradhvala, N. J., Huang, M. N., Tian Ng, A. W., Wu, Y., Boot, A., Covington, K. R., Gordenin, D. A., Bergstrom, E. N., Islam, S. M. A., Lopez-Bigas, N., Klimczak, L. J., McPherson, J. R., Morganella, S., Sabarinathan, R., Wheeler, D. A., Mustonen, V., Getz, G., Rozen, S. G., and Stratton, M. R.: The repertoire of mutational signatures in human cancer. Nature. 578(7793): 94–101, 2020.
- [93] Alexandrov, L. B., Kim, J., Haradhvala, N. J., Huang, M. N., Tian Ng, A. W., Wu, Y., Boot, A., Covington, K. R., Gordenin, D. A., Bergstrom, E. N., and others, : The repertoire of mutational signatures in human cancer. Nature. 578(7793): 94–101, 2020.
- [94] Seplyarskiy, V. B., Soldatov, R. A., Koch, E., McGinty, R. J., Goldmann, J. M., Hernandez, R. D., Barnes, K., Correa, A., Burchard, E. G., Ellinor, P. T., and others, : Population sequencing data reveal a compendium of mutational processes in the human germ line. Science. 373(6558): 1030–1035, 2021.

- [95] Wasser, S. K., Shedlock, A. M., Comstock, K., Ostrander, E. A., Mutayoba, B., and Stephens, M.: Assigning african elephant DNA to geographic region of origin: applications to the ivory trade. Proceedings of the National Academy of Sciences. 41: 14844–14852, 2004.
- [96] Falush, D., Stephens, M., and Pritchard, J. K.: Inference of population structure using multilocus genotype data: linked loci and correlated allele frequencies. Genetics. 164(4): 1567–1587, 2003.
- [97] Falush, D., Stephens, M., and J.K., P.: Inference of population structure using multilocus genotype data: dominant markers and null alleles. Molecular Ecology Notes. 7: 574–578, 2007.
- [98] Hubisz, M. J., Falush, D., Stephens, M., and Pritchard, J. K.: Inferring weak population structure with the assistance of sample group information. Molecular Ecology Resources. 9(5): 1322–1332, 2009.
- [99] Raj, A., Stephens, M., and Pritchard, J. K.: fastSTRUCTURE: variational inference of population structure in large SNP data sets. Genetics. 197(2): 573–589, 2014.
- [100] Kerimov, N., Hayhurst, J. D., Peikova, K., Manning, J. R., Walter, P., Kolberg, L., Samoviča, M., Sakthivel, M. P., Kuzmin, I., Trevanion, S. J., Burdett, T., Jupp, S., Parkinson, H., Papatheodorou, I., Yates, A. D., Zerbino, D. R., and Alasoo, K.: A compendium of uniformly processed human gene expression and splicing quantitative trait loci. Nature Genetics. 53(9): 1290–1299, 2021.
- [101] Wang, Q. S., Kelley, D. R., Ulirsch, J., Kanai, M., Sadhuka, S., Cui, R., Albors, C., Cheng, N., Okada, Y., Matsuda, K., Yamanashi, Y., Furukawa, Y., Morisaki, T., Murakami, Y., Kamatani, Y., Muto, K., Nagai, A., Obara, W., Yamaji, K., Takahashi, K., Asai, S., Takahashi, Y., Suzuki, T., Sinozaki, N., Yamaguchi, H., Minami, S., Murayama, S., Yoshimori, K., Nagayama, S., Obata, D., Higashiyama, M., Masumoto, A., Koretsune, Y., Aguet, F., Ardlie, K. G., MacArthur, D. G., and Finucane, H. K.: Leveraging supervised learning for functionally informed fine-mapping of cis-eQTLs identifies an additional 20,913 putative causal eQTLs. Nature Communications. 12: 3394, 2021.
- [102] Zhang, Z., Luo, K., Zou, Z., Qiu, M., Tian, J., Sieh, L., Shi, H., Zou, Y., Wang, G., Morrison, J., Zhu, A. C., Qiao, M., Li, Z., Stephens, M., He, X., and He, C.: Genetic analyses support the contribution of mRNA N6-methyladenosine (m6A) modification to human disease heritability. Nature Genetics. 52(9): 939–949, 2020.
- [103] Aracena, K. A., Lin, Y.-L., Luo, K., Pacis, A., Gona, S., Mu, Z., Yotova, V., Sindeaux, R., Pramatarova, A., Simon, M.-M., Chen, X., Groza, C., Loughheed, D., Gregoire, R., Brownlee, D., Li, Y., He, X., Bujold, D., Pastinen, T., Bourque, G., and Barreiro, L. B.: Epigenetic variation impacts ancestry-associated differences in the transcriptional response to influenza infection. bioRxiv. doi:10.1101/2022.05.10.491413, 2022.
- [104] Stark, R., Grzelak, M., and Hadfield, J.: RNA sequencing: the teenage years. Nature Reviews Genetics. 20(11): 631–656, November 2019. Number: 11 Publisher: Nature Publishing Group.
- [105] Minnoye, L., Marinov, G. K., Krausgruber, T., Pan, L., Marand, A. P., Secchia, S., Greenleaf, W. J., Furlong, E. E. M., Zhao, K., Schmitz, R. J., Bock, C., and Aerts, S.: Chromatin accessibility profiling methods. Nature Reviews Methods Primers. 1(1): 1–24, January 2021. Number: 1 Publisher: Nature Publishing Group.
- [106] Furey, T. S.: ChIP-seq and beyond: new and improved methodologies to detect and characterize protein–DNA interactions. Nature Reviews Genetics. 13(12): 840–852, December 2012. Number: 12 Publisher: Nature Publishing Group.
- [107] Zhang, Y., Shin, H., Song, J. S., Lei, Y., and Liu, X. S.: Identifying positioned nucleosomes with epigenetic marks in human from chip-seq. BMC Genomics. 9: 537, PMC2596141, 2008.
- [108] Feng, J., Liu, T., Qin, B., Zhang, Y., and Liu, X. S.: Identifying chip-seq enrichment using macs. Nature protocols. 7(9): 1728–1740, 2012.

- [109] Jaffe, A. E., Murakami, P., Lee, H., Leek, J. T., Fallin, M. D., Feinberg, A. P., and Irizarry, R. A.: Bump hunting to identify differentially methylated regions in epigenetic epidemiology studies. International Journal of Epidemiology. 41(1): 200–209, 2012.
- [110] Aryee, M. J., Jaffe, A. E., Corrada-Bravo, H., Ladd-Acosta, C., Feinberg, A. P., Hansen, K. D., and Irizarry, R. A.: Minfi: a flexible and comprehensive Bioconductor package for the analysis of Infinium DNA methylation microarrays. Bioinformatics. 30(10): 1363–1369, 2014.
- [111] Pedersen, B. S., Schwartz, D. A., Yang, I. V., and Kechris, K. J.: Comb-p: software for combining, analyzing, grouping and correcting spatially correlated P-values. Bioinformatics. 28(22): 2986–2988, 2012.
- [112] Lent, S., Xu, H., Wang, L., Wang, Z., Sarnowski, C., Hivert, M.-F., and Dupuis, J.: Comparison of novel and existing methods for detecting differentially methylated regions. BMC Genetics. 19: 84, 2018.
- [113] Yu, X. and Sun, S.: HMM-DM: identifying differentially methylated regions using a hidden Markov model. Statistical Applications in Genetics and Molecular Biology. 15(1): 69–81, 2016.
- [114] Shen, L., Zhu, J., Robert Li, S.-Y., and Fan, X.: Detect differentially methylated regions using non-homogeneous hidden Markov model for methylation array data. Bioinformatics. 33(23): 3701–3708, 2017.
- [115] Fernández, L., Pérez, M., Olanda, R., Orduña, J. M., and Marquez-Molins, J.: HPG-DHunter: an ultrafast, friendly tool for DMR detection and visualization. BMC Bioinformatics. 21: 287, 2020.
- [116] Gerard, D. and Stephens, M.: Empirical Bayes shrinkage and false discovery rate estimation, allowing for unwanted variation. Biostatistics. 21(1): 15–32, 2020.
- [117] Beagan, J. A. and Phillips-Cremens, J. E.: On the existence and functionality of topologically associating domains. Nature Genetics. 52(1): 8–16, January 2020.
- [118] Li, Y. I., Van De Geijn, B., Raj, A., Knowles, D. A., Petti, A. A., Golan, D., Gilad, Y., and Pritchard, J. K.: Rna splicing is a primary link between genetic variation and disease. Science. 352(6285): 600–604, 2016.
- [119] Shim, H. and Stephens, M.: Wavelet-based genetic association analysis of functional phenotypes arising from high-throughput sequencing assays. Annals of Applied Statistics. 9(2): 665–686, 2015.
- [120] Zheng, G. X. Y., Terry, J. M., Belgrader, P., Ryvkin, P., Bent, Z. W., Wilson, R., Ziraldo, S. B., Wheeler, T. D., McDermott, G. P., Zhu, J., Gregory, M. T., Shuga, J., Montesclaros, L., Underwood, J. G., Masquelier, D. A., Nishimura, S. Y., Schnall-Levin, M., Wyatt, P. W., Hindson, C. M., Bharadwaj, R., Wong, A., Ness, K. D., Beppu, L. W., Deeg, H. J., McFarland, C., Loeb, K. R., Valente, W. J., Ericson, N. G., Stevens, E. A., Radich, J. P., Mikkelsen, T. S., Hindson, B. J., and Bielas, J. H.: Massively parallel digital transcriptional profiling of single cells. Nature Communications. 8: 14049, 2017.
- [121] Ashburner, M., Ball, C. A., Blake, J. A., Botstein, D., Butler, H., Cherry, J. M., Davis, A. P., Dolinski, K., Dwight, S. S., Eppig, J. T., Harris, M. A., Hill, D. P., Issel-Tarver, L., Kasarskis, A., Lewis, S., Matese, J. C., Richardson, J. E., Ringwald, M., Rubin, G. M., and Sherlock, G.: Gene Ontology: tool for the unification of biology. Nature Genetics. 25(1): 25–29, 2000.
- [122] The Gene Ontology Consortium, : The Gene Ontology resource: enriching a Gold mine. Nucleic Acids Research. 49(D1): D325–D334, 2020.
- [123] Huang, D. W., Sherman, B. T., and Lempicki, R. A.: Systematic and integrative analysis of large gene lists using DAVID bioinformatics resources. Nature Protocols. 4: 44–57, 2009.
- [124] Jaakkola, T. S. and Jordan, M. I.: A variational approach to Bayesian logistic regression models and their extensions. In Madigan, D. and Smyth, P. (Eds.): Proceedings of the 16th International Workshop on Artificial Intelligence and Statistics. 1997, pp 283–294.

- [125] Zou, H. and Hastie, T.: Regularization and variable selection via the elastic net. Journal of the Royal Statistical Society, Series B. 67(2): 301–320, 2005.
- [126] Stephens, M., Dai, C., Lu, M., Gerard, D., Xiao, N., and Carbonetto, P. ashr: methods for adaptive Shrinkage, using empirical Bayes, 2016. R package version 2.0.5.
- [127] Efron, B.: Large-scale inference: empirical Bayes methods for estimation, testing, and prediction. Cambridge, UK, Cambridge University Press, 2010.
- [128] Dempster, A. P., Laird, N. M., and Rubin, D. B.: Maximum likelihood estimation from incomplete data via the EM algorithm. Journal of the Royal Statistical Society, Series B. 39: 1–38, 1977.
- [129] Carbonetto, P. and Stephens, M.: Integrated enrichment analysis of variants and pathways in genome-wide association studies indicates central role for IL-2 signaling genes in type 1 diabetes, and cytokine signaling genes in crohn’s disease. PLoS Genetics. 9(10): e1003770, 2013.
- [130] Wen, X., Pique-Regi, R., and Luca, F.: Integrating molecular QTL data into genome-wide genetic association analysis: Probabilistic assessment of enrichment and colocalization. PLoS Genetics. 13(3): e1006646, 2017.
- [131] Chen, X., Robinson, D. G., and Storey, J. D.: The functional false discovery rate with applications to genomics. Biostatistics. 22(1): 68–81, 2019.
- [132] Scott, J. G., Kelly, R. C., Smith, M. A., Zhou, P., and Kass, R. E.: False discovery rate regression: an application to neural synchrony detection in primary visual cortex. Journal of the American Statistical Association. 110(510): 459–471, 2015.
- [133] Boca, S. M. and Leek, J. T.: A direct approach to estimating false discovery rates conditional on covariates. PeerJ. 6: e6035, 2018.
- [134] Zablocki, R. W., Schork, A. J., Levine, R. A., Andreassen, O. A., Dale, A. M., and Thompson, W. K.: Covariate-modulated local false discovery rate for genome-wide association studies. Bioinformatics. 30(15): 2098–2104, 2014.
- [135] Linderman, S., Johnson, M. J., and Adams, R. P.: Dependent multinomial models made easy: stick-breaking with the Pólya-Gamma augmentation. Advances in Neural Information Processing Systems. 28, 2015.
- [136] Cerami, E. G., Gross, B. E., Demir, E., Rodchēnkov, I., Babur, Ö., Anwar, N., Schultz, N., Bader, G. D., and Sander, C.: Pathway Commons, a web resource for biological pathway data. Nucleic Acids Research. 39(suppl_1): D685–D690, 2011.
- [137] Rodchenkov, I., Babur, Ö., Luna, A. ., Aksoy, B. A., Wong, J. V., Fong, D., Franz, M., Šiper, M. C., Cheung, M., Wrana, M., Mistry, H., Mösier, L., Dlin, J., Wen, Q., O’Callaghan, C., Li, W. n., Elder, G., Smith, P. T., Dallago, C., Cerami, . E., Gross, B., Dogrusoz, U., Demir, E., Bader, G. D. ., and Sander, C.: Pathway Commons 2019 update: integration, analysis and exploration of pathway data. Nucleic Acids Research. 48(D1): D489–D497, 10 2019.
- [138] Liberzon, A., Birger, C., Thorvaldsdóttir, H., Ghandi, M., Mesirov, J. P., and Tamayo, P.: The molecular signatures database hallmark gene set collection. Cell Systems. 1(6): 417–425, 2015.
- [139] Liberzon, A., Subramanian, A., Pinchback, R., Thorvaldsdottir, H., Tamayo, P., and Mesirov, J. P.: Molecular signatures database (MSigDB) 3.0. Bioinformatics. 27(12): 1739–1740, 2011.
- [140] Geer, L. Y., Marchler-Bauer, A., Geer, R. C., Hăñ, L., He, J., He, S., Liu, C., Shi, W., and Bryănt, S. H.: The NCBI biosystems database. Nucleic Acids Research. 38(supplement-1): D492–D496, 10 2009.

- [141] Zhu, X. and Stephens, M.: Large-scale genome-wide enrichment analyses identify new trait-associated genes and pathways across 31 human phenotypes. Nature Communications. 9: 4361, 2018.
- [142] Journée, M., Nesterov, Y., Richtárik, P., and Sepulchre, R.: Generalized power method for sparse principal component analysis. Journal of Machine Learning Research. 11(15): 517–553, 2010.
- [143] Lazaridis, I., Patterson, N., Mitnik, A., Renaud, G., Mallick, S., Kirsanow, K., Sudmant, P. H., Schraiber, J. G., Castellano, S., Lipson, M., and others, : Ancient human genomes suggest three ancestral populations for present-day Europeans. Nature. 513(7518): 409–413, 2014.
- [144] Skoglund, P., Posth, C., Sirak, K., Spriggs, M., Valentin, F., Bedford, S., Clark, G. R., Reepmeyer, C., Petchey, F., Fernandes, D., and others, : Genomic insights into the peopling of the Southwest Pacific. Nature. 538(7626): 510–513, 2016.
- [145] Lipson, M., Sawchuk, E. A., Thompson, J. C., Oppenheimer, J., Tryon, C. A., Ranhorn, K. L., De Luna, K. M., Sirak, K. A., Olalde, I., Ambrose, S. H., and others, : Ancient DNA and deep population structure in sub-Saharan African foragers. Nature. 603(7900): 290–296, 2022.
- [146] Patterson, N., Moorjani, P., Luo, Y., Mallick, S., Rohland, N., Zhan, Y., Genschoreck, T., Webster, T., and Reich, D.: Ancient admixture in human history. Genetics. 192(3): 1065–1093, 2012.
- [147] Suvà, M. L. and Tirosh, I.: Single-cell RNA sequencing in cancer: lessons learned and emerging challenges. Molecular Cell. 75(1): 7–12, 2019.
- [148] Fan, J., Slowikowski, K., and Zhang, F.: Single-cell transcriptomics in cancer: computational challenges and opportunities. Experimental and Molecular Medicine. 52(9): 1452–1465, 2020.
- [149] Peng, J., Sun, B.-F., Chen, C.-Y., Zhou, J.-Y., Chen, Y.-S., others, , Zhao, Y., Yang, Y.-G., and Wu, W.: Single-cell RNA-seq highlights intra-tumoral heterogeneity and malignant progression in pancreatic ductal adenocarcinoma. Cell Research. 29(9): 725–738, 2019.
- [150] Chan-Seng-Yue, M., Kim, J. C., Wilson, G. W., Ng, K., and others, : Transcription phenotypes of pancreatic cancer are driven by genomic events during tumor evolution. Nature Genetics. 52(2): 231–240, 2020.
- [151] Guo, J. A., Hoffman, H. I., Weekes, C. D., Zheng, L., Ting, D. T., and Hwang, W. L.: Refining the molecular framework for pancreatic cancer with single-cell and spatial technologies. Clinical Cancer Research. 27(14): 3825–3833, 2021.
- [152] Raghavan, S., Winter, P. S., Navia, A. W., Williams, H. L., others, , Aguirre, A. J., and Shalek, A. K.: Microenvironment drives cell state, plasticity, and drug response in pancreatic cancer. Cell. 184(25): 6119–6137, 2021.
- [153] Townes, F. W., Hicks, S. C., Aryee, M. J., and Irizarry, R. A.: Feature selection and dimension reduction for single-cell rna-seq based on a multinomial model. Genome biology. 20(1): 1–16, 2019.
- [154] Landa, B., Zhang, T. T., and Kluger, Y.: Biwhitening reveals the rank of a count matrix. arXiv preprint arXiv:2103.13840. 2021.
- [155] van der Maaten, L. and Hinton, G.: Visualizing data using t-SNE. Journal of Machine Learning Research. 9(86): 2579–2605, 2008.
- [156] Becht, E., McInnes, L., Healy, J., Dutertre, C.-A., Kwok, I. W., Ng, L. G., Ginhoux, F., and Newell, E. W.: Dimensionality reduction for visualizing single-cell data using umap. Nature biotechnology. 37(1): 38–44, 2019.
- [157] Auton, A., Abecasis, G. R., Altshuler, D. M., Durbin, R. M., and others, : A global reference for human genetic variation. Nature. 526(7571): 68–74, 2015.

- [158] Zerjal, T., Dashnyam, B., Pandya, A., Kayser, M., Roewer, L., Santos, F. R., Schiefenhövel, W., Fretwell, N., Jobling, M. A., Harihara, S., Shimizu, K., Semjidsmaa, D., Sajantila, A., Salo, P., Crawford, M. H., Ginter, E. K., Evgrafov, O. V., and Tyler-Smith, C.: Genetic relationships of Asians and Northern Europeans, revealed by Y-chromosomal DNA analysis. American Journal of Human Genetics. 60(5), 1997.
- [159] Van der Maaten, L. and Hinton, G.: Visualizing data using t-sne. Journal of machine learning research. 9(11), 2008.
- [160] Moffitt, R. A., Marayati, R., Flate, E. L., Volmar, K. E., Loeza, S. G. H., Hoadley, K. A., Rashid, N. U., Williams, L. A., Eaton, S. C., Chung, A. H., and others, : Virtual microdissection identifies distinct tumor- and stroma-specific subtypes of pancreatic ductal adenocarcinoma. Nature Genetics. 47(10): 1168–1178, 2015.
- [161] Collisson, E. A., Bailey, P., Chang, D. K., and Biankin, A. V.: Molecular subtypes of pancreatic cancer. Nature reviews Gastroenterology & hepatology. 16(4): 207–220, 2019.
- [162] Hwang, W. L., Jagadeesh, K. A., Guo, J. A., Hoffman, H. I., Yadollahpour, P., Reeves, J. W., Mohan, R., Drokhyansky, E., Van Wittenberghe, N., Ashenberg, O., and others, : Single-nucleus and spatial transcriptome profiling of pancreatic cancer identifies multicellular dynamics associated with neoadjuvant treatment. Nature Genetics. 54(8): 1178–1191, 2022.
- [163] Bailey, P., Chang, D. K., Nones, K., Johns, A. L., Patch, A.-M., Gingras, M.-C., Miller, D. K., Christ, A. N., Bruxner, T. J., Quinn, M. C., and others, : Genomic analyses identify molecular subtypes of pancreatic cancer. Nature. 531(7592): 47–52, 2016.
- [164] Raphael, B. J., Hruban, R. H., Aguirre, A. J., Moffitt, R. A., Yeh, J. J., Stewart, C., Robertson, A. G., Cherniack, A. D., Gupta, M., Getz, G., and others, : Integrated genomic characterization of pancreatic ductal adenocarcinoma. Cancer Cell. 32(2): 185–203, 2017.
- [165] Aung, K. L., Fischer, S. E., Denroche, R. E., Jang, G.-H., Dodd, A., Creighton, S., Southwood, B., Liang, S.-B., Chadwick, D., Zhang, A., and others, : Genomics-driven precision medicine for advanced pancreatic cancer: early results from the COMPASS trial. Clinical Cancer Research. 24(6): 1344–1354, 2018.
- [166] Cao, L., Huang, C., Zhou, D. C., Hu, Y., Lih, T. M., Savage, S. R., Krug, K., Clark, D. J., Schnaubelt, M., Chen, L., and others, : Proteogenomic characterization of pancreatic ductal adenocarcinoma. Cell. 184(19): 5031–5052, 2021.
- [167] Shah, V. M., Sheppard, B. C., Sears, R. C., and Alani, A. W.: Hypoxia: Friend or foe for drug delivery in pancreatic cancer. Cancer letters. 492: 63–70, 2020.
- [168] Tao, J., Yang, G., Zhou, W., Qiu, J., Chen, G., Luo, W., Zhao, F., You, L., Zheng, L., Zhang, T., and others, : Targeting hypoxic tumor microenvironment in pancreatic cancer. Journal of Hematology & Oncology. 14(1): 1–25, 2021.